A Project Report

On

Deep Autoencoder-Based Denoising of Dynamic PET Scans

Submitted in partial fulfillment of the requirements
for the award of the degree in

MASTER OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

(Communication and Signal Processing)

by

M NARENDRA KUMAR

20011P0413

Under the esteemed guidance of

Dr. A.RAJANI

Professor

ECE Department, JNTUH-UCESTH

Department of Electronics and Communication Engineering

Jawaharlal Nehru Technological University Hyderabad -University College of Engineering, Science & Technology Hyderabad

(Kukatpally, Hyderabad-500 085, Telangana)

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Jawaharlal Nehru Technological University Hyderabad-University College of Engineering, Science & Technology Hyderabad

(Kukatpally, Hyderabad – 500085, Telangana)

Department of Electronics and Communication Engineering

DECLARATION BY THE CANDIDATE

I, **M. Narendra Kumar(20011P0413),** hereby declare that the major project entitled ***"Deep Autoencoder-Based Denoising of Dynamic PET Scans"***, under the guidance of **Dr. A. Rajani** is submitting in the partial fulfillment of the requirements for the award of the degree in **Master of Technology** in **Communication and Signal Processing, Department of Electronics and Communication Engineering .** This is a record of bonafide work carried out by me during the academic year 2024-2025. The results embodied in this project have not been reproduced or copied from any source, nor submitted to any other University or Institute for the award of any other degree or diploma.

M. Narendra Kumar

20011P0413

Jawaharlal Nehru Technological University Hyderabad University
College of Engineering, Science & Technology Hyderabad

(Kukatpally, Hyderabad – 500085, Telangana)

Department of Electronics and Communication Engineering

CERTIFICATE BY SUPERVISOR

This is to certify that the Major Project entitled ***"Deep Autoencoder-Based Denoising of Dynamic PET Scans"*** being submitted by **M. Narendra Kumar (20011P0413)** in the partial fulfillment of the requirements for the award of the degree of **Master of Technology** in **Communication and Signal Processing, Department of Electronics and Communication Engineering** at **Jawaharlal Nehru Technological University Hyderabad** during the academic year 2024-2025 is a record of bonafide work carried out by him under my guidance and supervision. The results have been verified and found to be satisfactory.

Project Guide

Dr. A. Rajani

Professor,

Department of ECE

JNTUHUCESTH

Jawaharlal Nehru Technological University Hyderabad University College of Engineering, Science & Technology Hyderabad

(Kukatpally, Hyderabad – 500085, Telangana)

Department of Electronics and Communication Engineering

CERTIFICATE BY THE HEAD OF THE DEPARTMENT

This is to certify that the Major Project entitled ***"Deep Autoencoder-Based Denoising of Dynamic PET Scans"*** being submitted by **M. Narendra Kumar (20011P0413)** in the partial fulfillment of the requirements for the award of the degree of **Master of Technology** in **Communication and Signal Processing, Department of Electronics and Communication Engineering** at **Jawaharlal Nehru Technological University Hyderabad** is a record of bonafide work carried out by him during the academic year 2024-2025.

Head of the Department

Dr. T. Madhavi Kumari

Associate Professor and Head

Department of ECE

JNTUHUCESTH

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M. Narendra Kumar

20011P0413

ABSTRACT

In the field of medical imaging, dynamic PET (Positron Emission Tomography) shows promise due to its capability to capture temporal variations in physiological processes. However, the presence of noise often degrades these images, hindering their diagnostic effectiveness. To tackle this issue, a novel approach that employs a Tracer-Specific Deep Artificial Neural Network (DNN) is introduced. This technique takes advantage of the distinct characteristics of PET tracers to effectively reduce noise in dynamic PET images, enhancing both spatial resolution and the signal-to-noise ratio. By integrating advanced machine learning methods with specialized knowledge about tracer behavior, this DNN-based strategy presents a robust solution to improve the quality and dependability of dynamic PET imaging, thereby enabling more accurate clinical evaluations and advancing research in neurology, oncology, and cardiology.

Present-day methods for denoising dynamic PET images mainly depend on traditional filtering techniques, such as Gaussian smoothing or iterative reconstruction algorithms. Although these approaches can mitigate noise, they frequently sacrifice spatial resolution or introduce artifacts, thus limiting their efficacy. Furthermore, they typically do not fully utilize the specific attributes of PET tracers and their dynamic behaviours. This has led to the investigation of more sophisticated denoising methods customized to the unique characteristics of PET tracers, with deep learning presenting promising enhancements.

Our approach utilizes deep artificial neural networks (DNNs) that are specifically designed for PET tracers to improve the quality of dynamic PET images. By training on a comprehensive dataset of tracer-specific dynamic PET images, the DNN grasps the intricate relationships between temporal dynamics and tracer uptake patterns. This allows for effective noise suppression while maintaining fine details and spatial accuracy. Moreover, the proposed system lowers the computational load compared to iterative techniques, making it suitable for real-time use in clinical environments. This method signifies a notable progression in dynamic PET imaging, optimizing image quality and aiding in more precise diagnostic and therapeutic choices.

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CHAPTER-1
INTRODUCTION

Positron Emission Tomography (PET) imaging serves as an essential diagnostic instrument in contemporary medical research and clinical practice, providing vital insights into physiological functions and the advancement of diseases. Nonetheless, this technique confronts a fundamental issue: achieving the right balance between limiting patient exposure to radiation and ensuring the production of high-quality diagnostic images. Existing PET scanning protocols frequently necessitate a compromise between lowering the doses of radioactive tracers and the quality of images, where reduced doses typically yield significantly noisier images that can conceal important diagnostic information.

This research tackles this crucial constraint by introducing an innovative method for image denoising utilizing Deep Autoencoder (DAE) technology. By creating a deep learning model tailored to specific tracers, we seek to enhance the interpretability of low-dose PET scans while preserving the highest feasible level of diagnostic accuracy. Our approach is notably important as it addresses the intrinsic Poisson noise linked with PET imaging, providing a focused resolution that surpasses standard denoising techniques.

The methodology we propose not only aims to diminish the radiation risk faced by patients but also holds promise for improving diagnostic consistency across various medical imaging environments. By safeguarding key structural and functional information while efficiently eliminating noise, this technique has the potential to transform how healthcare professionals analyze dynamic PET images, especially in long-term research and precision diagnostics.

The main innovation of this study resides in its application of machine learning, which employs deep learning algorithms specifically crafted to recognize and reduce the unique noise challenges present in PET imaging. By training our Deep Autoencoder on an extensive dataset of PET scans, we construct an advanced neural network capable of differentiating between relevant biological signals and noise caused by artifacts. This approach transcends conventional signal processing methods, presenting a more intelligent and adaptable answer to image denoising. Our technique carries significant implications for demanding diagnostic situations, such as the early detection of tumors, monitoring the progression of neurological diseases, and conducting intricate metabolic studies where subtle details in images may be crucial for precise interpretation.

1.1 LITERATURE REVIEW

Denoising in dynamic Positron Emission Tomography (PET) imaging is critical due to the inherently low signal-to-noise ratio (SNR) resulting from limited photon counts per time frame. Traditional filtering methods often compromise spatial or temporal resolution. Recent advances in deep learning, particularly autoencoder-based models, have shown significant promise in addressing these challenges.

Klyuzhin et al. [1] introduced a tracer-specific Deep Autoencoder (DAE) trained on paired noisy and noise-free image patches from [11C] raclopride PET data. Their model significantly reduced voxel-level noise while preserving fine anatomical details, outperforming traditional denoising methods such as Gaussian smoothing, HYPR, and STEM. This study demonstrated that DAEs, when trained on representative datasets, can yield substantial improvements in structural similarity and SNR without introducing bias.

A widely studied traditional method is HYPR-LR (Highly Constrained Back Projection with Local Reconstruction), proposed by Christian et al. [2]. This technique constructs a high-SNR composite image to guide the enhancement of individual time frames. Applied to phantom, animal, and human studies, HYPR-LR improved image quality and quantitative accuracy while preserving spatial resolution, marking a significant step forward in low-SNR PET imaging.

Building upon deep learning foundations, Kim et al. [3] proposed a CNN-based denoising framework that does not rely on any pre-existing training dataset. Their model utilizes temporal redundancy across frames to suppress noise, improving PSNR and detail retention. This approach is particularly suited to clinical applications where acquiring clean ground-truth data is infeasible.

In another innovative direction, Eo et al. [4] developed a hybrid model combining Deep Image Prior (DIP) and Regularization by Denoising (RED). This unsupervised strategy leverages the structural prior inherent in convolutional networks along with pretrained denoisers. The model enhances both spatial and temporal fidelity, outperforming traditional and supervised deep learning methods without requiring labelled data.

Li et al. [5] introduced a DL-based Joint Filtering (DLJF) approach using co-registered MRI as guidance. Their dual-branch CNN with attention-based fusion improved SNR and spatial detail, even without clean PET labels.

Focusing on classical statistical approaches, Dutta et al. [6] proposed a spatiotemporal Non-Local Means (NLM) method tailored for dynamic PET imaging. Their adaptive framework applies local variance-based smoothing to reduce noise while preserving contrast. Validation on both simulated and real datasets revealed superior bias-variance performance over conventional methods including Gaussian, PCA, HYPR, and traditional NLM.

In the broader scope of image reconstruction, Toyonaga et al. [7] introduced a deep learning–based attenuation correction model for PET images. Utilizing MLAA-generated maps and a U-Net architecture with physics-informed loss, the model effectively generated CT-equivalent attenuation maps. This improved both image quality and quantitative tumor assessment across tracers such as 18F-FDG, 68Ga-DOTATATE, and 18F-Fluciclovine, highlighting the importance of integrating denoising with end-to-end imaging pipelines.

Finally, Hashimoto et al. [8] provided a comprehensive review of deep learning techniques in PET image denoising and reconstruction. They traced the evolution from traditional reconstruction to deep learning–based post-processing, direct mapping, and iterative hybrid models. Their survey also explored emerging architectures such as GANs, transformers, and DIP-based models, and emphasized the integration of these algorithms with cutting-edge PET hardware including total-body scanners and time-of-flight (TOF) systems.

Together, these studies highlight a clear trajectory in PET denoising research—from hand-engineered filters to sophisticated deep learning architectures that exploit spatial, temporal, and anatomical priors. In particular, deep autoencoders and unsupervised neural frameworks hold substantial promise for delivering real-time, high-fidelity dynamic PET reconstructions in clinical and research settings.

1.2 MOTIVATION

Positron Emission Tomography (PET) is a highly effective imaging technique commonly utilized for clinical diagnostics, especially in fields like oncology, neurology, and cardiology. Nevertheless, the radioactive tracers employed in PET scans subject patients to ionizing radiation, which raises safety issues, particularly for children and in cases involving multiple scans. It is desirable to minimize radiation exposure, but doing so leads to low-count images that are noisy and more challenging to interpret. The aim is to uphold image quality even when the tracer dose is diminished. This project seeks to tackle that problem by applying deep learning to denoise low-dose PET images without sacrificing diagnostic accuracy.

Conventional denoising methods, such as Gaussian filtering or Non-Local Means, often result in a blurring of fine structural details and tend to overlook dynamic changes within the image sequences. Deep Autoencoders, however, present a compelling alternative by learning high-level features and reconstructing clear images from noisy inputs. By training on synthetic or low-dose datasets, the autoencoder becomes adept at learning the intricate mappings required to reduce noise while preserving vital anatomical and functional details. The project utilizes a 2.5D Deep Autoencoder framework, enabling it to factor in both spatial and temporal correlations in dynamic PET sequences. This improves the capability to eliminate noise while retaining temporal consistency across the frames of the images.

This research has the potential to significantly enhance patient outcomes by facilitating high-quality PET imaging at lower radiation doses, thereby promoting safer and more frequent imaging when necessary. Furthermore, it could help decrease healthcare expenses by enhancing scan efficiency and lowering the necessity for repeat imaging due to subpar image quality. Additionally, precise denoising may improve quantitative metrics such as Standardized Uptake Value (SUV), which is crucial for monitoring treatment effectiveness and tracking disease progression. Integrating these deep learning-driven denoising techniques into the PET imaging workflow could enable real-time processing, thereby supporting faster diagnoses. Overall, the project aims to make PET imaging more accessible, dependable, and patient-centered.

The inspiration for this project stems from the necessity for safer, more precise, and efficient PET imaging within clinical environments. The specific objectives of this project include:

- Reducing the radiation dosage during PET scans while ensuring high image quality to enhance patient safety.
- Improving the clarity of low-dose dynamic PET images through deep learning methods rather than traditional filtering approaches.
- Preserving intricate anatomical features and maintaining temporal consistency across image sequences to ensure reliable clinical interpretation.
- PET imaging is more reachable and cost-effective by lessening the requirement for repeat scans due to inadequate image quality.

1.3 AIM:

The objective of this project is to create a deep autoencoder (DAE) model for the purpose of denoising PET (Positron Emission Tomography) images, which will improve image quality and decrease radiation exposure to patients by allowing for scans with lower doses.

1.4 OBJECTIVES:

- To import DICOM PET data, arrange it into series, construct 3D volumes, and standardize the image data.
- To generate synthetic low-dose PET images by introducing noise (Poisson, Gaussian, Motion Artifacts), extract 3D patches, and create training datasets consisting of paired noisy and clean images.
- To create and train a deep autoencoder neural network with several hidden layers aimed at understanding noise patterns and restoring clean images.
- To utilize the trained model for denoising test volumes, compute quality metrics (PSNR, MSE, SSIM, Brisque Values), and produce comparative visualizations (slices, difference maps) for performance evaluation.

1.5 BLOCK DIAGRAM:

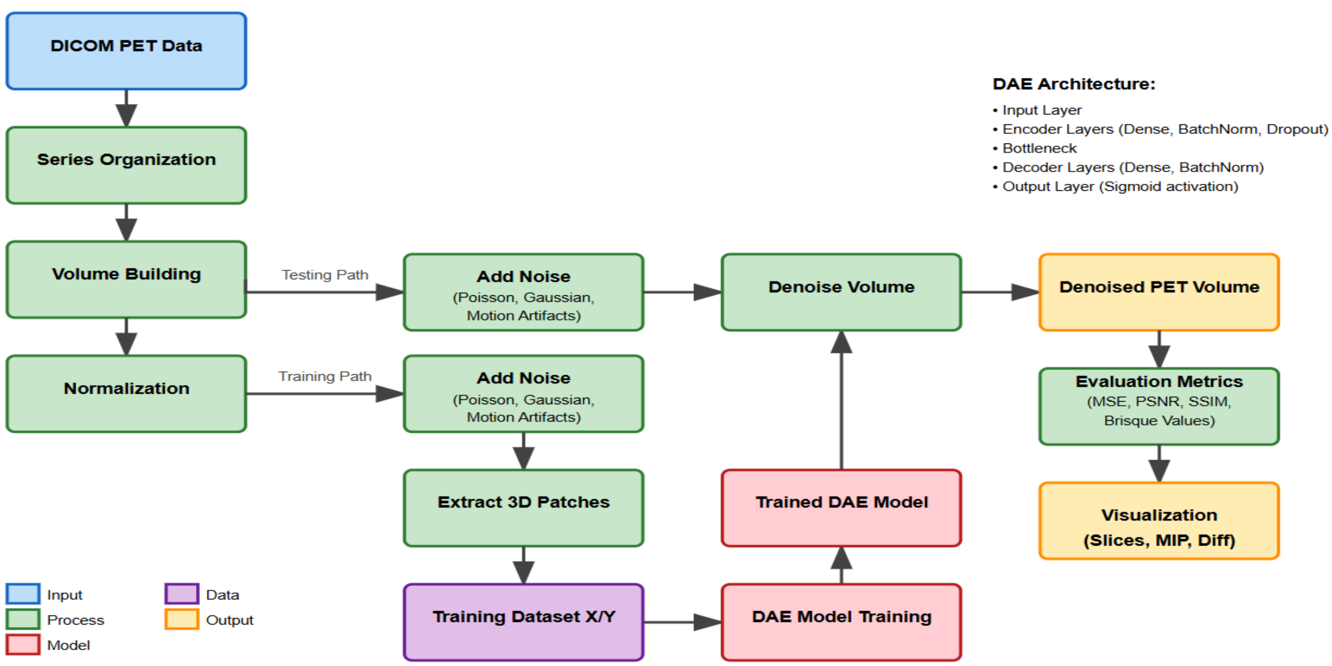


Figure 1.1 Block Diagram

The workflow for PET image denoising initiates with separate 2D DICOM files grouped into series according to their SeriesInstanceUID, facilitating proper spatial and temporal alignment. These 2D slices are subsequently combined to form comprehensive 3D volumes that depict the entirety of the patient's scan data, followed by normalization to adjust pixel values to a uniform range of 0-1. This preprocessing step guarantees consistent input across variations in scanners or acquisition parameters, laying a reliable groundwork for the subsequent denoising processes. To generate effective training datasets, the pipeline intentionally adds simulated noise (including Poisson, Gaussian, or motion artifacts) to clean images, creating paired datasets of noisy and clean volumes. These volumes are then segmented into smaller 3D patches, which function as individual training samples, with corresponding pairs of noisy and clean patches input into the Deep Autoencoder (DAE) model. The DAE design includes encoder layers that compact the noisy input, a bottleneck that retains vital features, and decoder layers that reconstruct the clean image, progressively learning to translate from degraded to flawless representations.

After the training process, the DAE model is utilized to denoise complete PET volumes by handling overlapping patches and seamlessly merging the outcomes. The quality of the denoised output is thoroughly assessed using metrics such as MSE, PSNR, SSIM, and Brisque, which measure the enhancement in image quality against ground truth data. Lastly, visualization methods display the denoised volumes via axial, coronal, and sagittal slice views, enabling clinicians to confirm that the denoising procedure has effectively preserved essential anatomical and functional details while successfully eliminating noise that might otherwise hinder diagnostic reliability.

1.6 DISSERTATION ORGANIZATION

This dissertation is structured into six chapters, each focusing on distinct aspects from fundamental principles to methodology, execution, and prospective advancements.

Chapter 1: Introduction – This chapter offers a summary of the dissertation, outlining the motivation for employing deep autoencoders to denoise dynamic PET scans, the primary goal and objectives, and a concise overview of the suggested denoising framework. Additionally, it includes a literature review discussing previous studies on PET image denoising and the application of deep learning in medical imaging.

Chapter 2: Denoising Techniques for PET Images – This chapter surveys various denoising methods utilized in PET imaging. It examines both traditional filtering and reconstruction-based techniques, as well as recent progress in machine learning and deep learning, emphasizing their benefits and drawbacks.

Chapter 3: Dataset and Preprocessing – This chapter describes the dataset used for this research, including its origins, features, and challenges. It also addresses noise modeling in dynamic PET scans, techniques for patch extraction, and normalization processes implemented to prepare the data for the deep autoencoder.

Chapter 4: Methodology – This chapter outlines the architecture of the proposed Deep Autoencoder (DAE) model. It provides details about the encoder, bottleneck, and decoder layers, the activation functions used, and how patch-based inputs are handled. Each design choice and its reasoning are discussed comprehensively.

Chapter 5: Implementation and Results – This chapter showcases the practical execution of the proposed denoising model. It describes the training setup, evaluates metrics, provides a comparative analysis against baseline methods, and presents both visual and quantitative outcomes that demonstrate the effectiveness of the deep autoencoder in mitigating noise in dynamic PET images.

Chapter 6: Conclusion and Future Scope – This chapter recaps the essential findings and evaluates the performance of the proposed denoising framework. It also highlights possible future research avenues, such as investigating alternative architectures, incorporating temporal information in dynamic scans, and improving computational efficiency.

CHAPTER-2

DENOISING TECHNIQUES FOR PET IMAGES

PET image denoising refers to the technique of eliminating unwanted random fluctuations (noise) from Positron Emission Tomography (PET) images to enhance their clarity and precision. PET imaging illustrates the functioning of organs and tissues by detecting radioactive tracers; however, these images frequently exhibit noise caused by low radiation exposure, limited scanning duration, or the statistical nature of radiation detection.

The benefits include:

1. Improved diagnosis - Clearer images enable physicians to identify diseases sooner and with greater accuracy.
2. Patient safety - Facilitates the use of reduced radiation doses while preserving image quality.
3. Quicker scans - Minimizes the scan duration needed, increasing patient comfort.
4. Treatment planning - Provides more accurate data for surgical and radiation therapy procedures.
5. Cost efficiency - Decreases the necessity for repeat scans due to subpar image quality.

2.1 Different PET Image Denoising Techniques:

1. Gaussian Filtering:

This approach softens the image by averaging each pixel with its surrounding neighbors using a bell-shaped curve. It resembles applying a gentle focus effect that smooths out random noise while diminishing edge sharpness. Although it's straightforward and quick, it may eliminate crucial details along with the noise. This method is most effective when the noise is uniformly spread across the image.

2. Wavelet Transform Denoising:

This method decomposes the image into various frequency components (similar to differentiating musical notes into high and low sounds) and cleans out noise from specific frequency bands. It is superior to basic blurring techniques in keeping edges and significant features intact. The approach is based on the premise that valuable image information and noise occupy different frequency ranges. Optimal results require careful selection of wavelet types and threshold values.

3. Total Variation (TV) Denoising:

This technique aims to achieve smooth areas while maintaining distinct edges by minimizing the overall variation in the image. It works by penalizing quick changes in pixel values, except at actual edges where such changes are expected. The method is particularly effective in preserving boundaries between various organs or tissues. However, it can lead to a "staircase" effect where gradual gradients appear as flat sections.

4. Non-Local Means Denoising:

Rather than simply examining nearby pixels, this technique scans the entire image for similar patterns and utilizes them to estimate the accurate pixel value. It is based on the understanding that natural images feature repetitive structures and textures. This strategy performs well for images with recurring patterns, although it can be computationally intensive. It is more effective than local methods when the image contains self-similar structures.

5. Deep Learning Autoencoders:

These are neural networks designed to compress and reconstruct images, learning to distinguish noise from relevant information during the training process. The network is trained using pairs of noisy and clean images to understand the denoising method. Autoencoders can capture intricate noise patterns and image features that conventional techniques might overlook. They adjust to the unique characteristics of the dataset they are trained on.

2.2 Disadvantages of Other Methods over Deep Autoencoders:

Gaussian Filtering:

It removes significant details and edges along with noise, failing to differentiate between genuine image features and noise. The uniform approach does not adapt to varying regions within the image.

Wavelet Transform:

It necessitates the manual selection of wavelet type and parameters, with its effectiveness largely reliant on the selected thresholds. Handling complex, non-stationary noise may pose challenges for this method.

Total Variation:

This technique can introduce artificial "staircase" effects in smooth areas and may excessively smooth out fine textures and details. It demands considerable computational resources for large images and requires careful adjustment of parameters.

Non-Local Means:

It is very slow when processing large images, operating under the assumption that images feature self-similar patterns. This method might struggle with unique or non-repetitive patterns and requires significant memory for patch matching.

2.3 Advantages of Choosing Autoencoders for Denoising

Adaptive Learning:

Autoencoders autonomously identify the most effective method for eliminating noise from your particular image types, rather than relying on rigid rules that may not be suitable for every situation.

Feature Preservation:

They differentiate between critical image components (such as the boundaries of organs) and noise, safeguarding essential diagnostic details while removing irrelevant artifacts.

End-to-End Optimization:

The entire denoising procedure is jointly optimized throughout the training phase, resulting in a more integrated and efficient solution compared to approaches that involve multiple distinct steps.

Handling Complex Noise:

They can learn to eliminate intricate noise patterns that conventional techniques find challenging, including noise that varies spatially and artifacts unique to PET imaging.

Scalability:

Once trained, they can quickly process images and efficiently manage large datasets, making them suitable for clinical applications.

Customization:

They can be trained on specific PET scans (pertaining to different organs, tracers, or types of scanners) to deliver tailored denoising for specific purposes.

No Parameter Tuning:

In contrast to traditional techniques that necessitate manual fine-tuning of various parameters, autoencoders automatically determine the best parameters during the training process.

CHAPTER-3

DATASET AND PREPROCESSING

3.1 Dataset:

The dataset utilized in this research comprises 4D DICOM chest PET scan images acquired from Turun Yliopistollinen sairaala (Turku University Hospital) in Turku, Finland, using a GE Medical Systems Discovery MI PET scanner (station name: "Aino"), making it well-suited for dynamic PET imaging analysis and denoising investigations due to its standardized acquisition protocols and high-quality annotations.

This dataset is organized into 24 sub-datasets, with each containing 1704 images that reflect a variety of imaging conditions. These sub-datasets are classified according to important parameters such as radiotracer dosage percentages (for instance, 20% and 60%) and energy windows (for example, 200–160 eV or 250–100 eV), indicated in naming conventions such as PT_20p 200_160 BSREM or PT_60p 250_100 BSREM. The PET images are reconstructed using two frequently employed algorithms: OSEM (Ordered Subset Expectation Maximization) and BSREM (Block Sequential Regularized Expectation Maximization). OSEM is known for its swift convergence, while BSREM offers improved noise reduction without compromising structural details.

The scans were conducted following a standard chest PET/CT protocol using H2O (water) labeled with Oxygen-15 (^15^O) as the radiotracer, with the protocol name 5.22 PTCT_RGD_H2O_LEPO. The study was performed on May 10, 2021, with radiotracer activity of 500 MBq (megabecquerels) and a half-life of 122.24 seconds, administered at 13:17:44. Each PET slice has a matrix size of 192×192, a slice thickness of 3.75 mm. The imaging coverage extends from -89.6 mm to 105.7 mm with a total range of 195.3 mm, focusing on the thorax region with particular emphasis on cardiac imaging. Patients were positioned in feet-first supine (FFS) orientation during acquisition.

This appears to be a research dataset from a phantom study (Patient ID: 100521-sip15, Name: SIP15^Phantom_Aino), indicating controlled experimental conditions rather than human patient data. The use of O-15 water PET imaging makes it particularly suitable for perfusion or blood flow studies. The diverse levels of tracer activity and reconstruction methods create an optimal environment for developing and assessing denoising algorithms, particularly deep autoencoders, by mimicking low-dose imaging and a range of noise characteristics.

3.2 Preprocessing – Noise Modeling:

To create realistic noisy conditions for training, noise modeling was implemented on the normalized PET volumes. This involves adding Poisson noise (which represents photon counting statistics), Gaussian noise (which simulates system and thermal noise), and motion artifacts (to replicate patients' movements during image acquisition). These intentionally degraded images are used as inputs for the denoising autoencoder, enabling the model to learn effective noise-removal techniques across different types of degradation and various imaging scenarios.

1. Poisson Noise - The Quantum Challenge

Poisson noise is a fundamental statistical limitation inherent in PET imaging, stemming from the quantum characteristics of photon detection and the random nature of radioactive decay. Unlike other imaging techniques, PET depends on the detection of random pairs of gamma-ray photons created during positron-electron annihilation events, rendering this noise unavoidable and closely tied to the physics behind the imaging process. The impact of Poisson noise is inversely related to the count of detected events, indicating that scans performed with lower doses or areas with minimal tracer uptake experience relatively greater noise levels.

$$P(X = k) = \frac{\lambda^k \cdot e^{-\lambda}}{k!}$$

Where:

- $P(X = k)$ is the probability of observing k photon counts,
- λ (lambda) is the expected count rate or mean photon count,
- e is Euler’s number (approximately 2.718),
- k! is the factorial of k.

The practical application of simulating Poisson noise in PET denoising studies involves scaling the clean image based on a signal-to-noise ratio (SNR) factor, carrying out Poisson random sampling, and subsequently rescaling to the original intensity range. This method effectively replicates the connection between signal intensity and noise variance that is typical in real PET scans. The selection of SNR=5 in common implementations signifies a moderately noisy condition that resembles low-dose clinical protocols aimed at minimizing radiation exposure for patients.

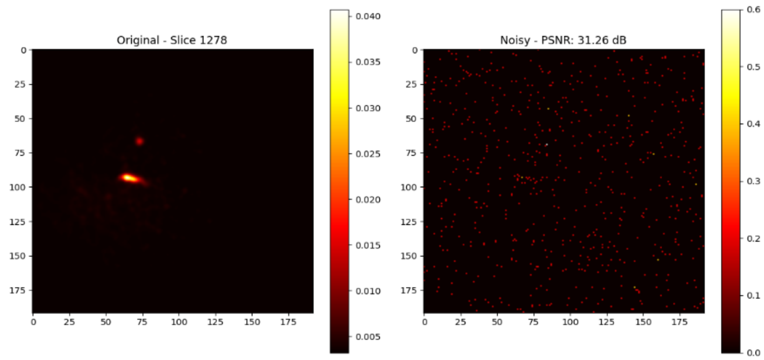
Key Parameters and Effects:

SNR = 5: Indicates a moderate level of noise, reflective of low-dose protocols

SNR = 10: Signifies reduced noise, typical of standard clinical acquisitions

SNR = 2: Represents high noise, associated with ultra-low-dose or very brief scanning durations

Spatial Distribution: Non-uniform, with increased noise levels in areas of low uptake



Original Image | Image with Poisson Noise

Fig 3.1: Slices view under Poisson Noise

2. Gaussian Noise - Electronic System Interference

Gaussian noise present in PET imaging arises from the electronic components within the detection system, such as photomultiplier tubes, analog-to-digital converters, and signal processing circuits. This additive white noise adheres to a normal distribution and is not influenced by the intensity of the signal, which differentiates it from the signal-dependent characteristics of Poisson noise. Although modern PET scanners have greatly enhanced the qualities of electronic noise, it still poses a challenge in areas with low counts and can combine with Poisson noise, leading to a decline in overall image quality. This particularly impacts the ability to detect small lesions and accurately measure tracer uptake.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

- **μ = 0.0** (zero mean),

- **σ = 0.05** (standard deviation),
- **f(x)** represents the probability density function of the Gaussian distribution,
- **x** is the random variable

The process of simulating Gaussian noise involves generating random samples from a normal distribution characterized by a mean of zero and a specified standard deviation, and then directly incorporating these values into the clean image data. The selection of $\sigma = 0.05$ indicates approximately 5% noise in relation to the normalized intensity range of the image, which reflects the performance of contemporary PET scanners. In contrast to Poisson noise, Gaussian noise uniformly impacts all pixels irrespective of their intensity levels, making it particularly prominent in background areas and regions with consistent tracer distribution.

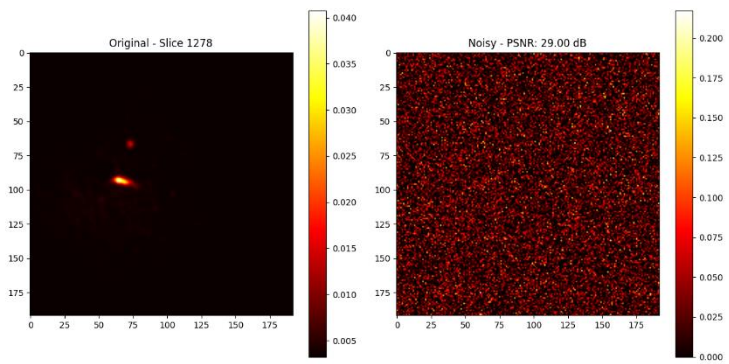
Characteristics of Noise:

Mean ($\mu = 0$): Guarantees that there is no systematic deviation in intensity values.

Standard Deviation ($\sigma = 0.05$): Represents a 5% noise level pertaining to the signal range.

Distribution: Symmetrical around zero, with 68% of values falling within $\pm 1\sigma$.

Spatial Pattern: Evenly distributed across the whole image.



Original Image Image with Gaussian Noise

Fig 3.2: Slices view under Gaussian Noise

3. Motion Artifacts - Patient Movement Effects

Motion artifacts in PET imaging arise from unavoidable patient movements during the relatively lengthy acquisition times (ranging from 15 to 30 minutes for whole-body scans), leading to spatial misregistration among various temporal segments of the scan. These artifacts appear as blurring, inconsistencies between adjacent slices, and geometric distortions that can greatly affect both visual assessment and quantitative analysis. Patient movement poses a significant challenge, especially in elderly individuals, children, and those with conditions that hinder prolonged stillness, such as chronic pain or respiratory issues.

$$I_{\text{motion}}(x, y, z) = I(x + \Delta x,\ y + \Delta y,\ z)$$

- Here, motion shifts the image/volume in **x and y directions** by Δx and Δy.
- The **z coordinate remains unchanged** (no shift along z).

The simulation of motion artifacts consists of applying random spatial displacements to individual slices with a defined probability, generating realistic discontinuities between adjacent slices that replicate the effects of genuine patient movement. The method uses a 20% probability for each slice and allows for maximum shifts of 2 pixels, reflecting moderate patient motion that is typical in clinical situations. This technique accurately represents the inconsistencies between slices that occur when patients move during different phases of imaging acquisition, which results in anatomical structures appearing misaligned across successive slices.

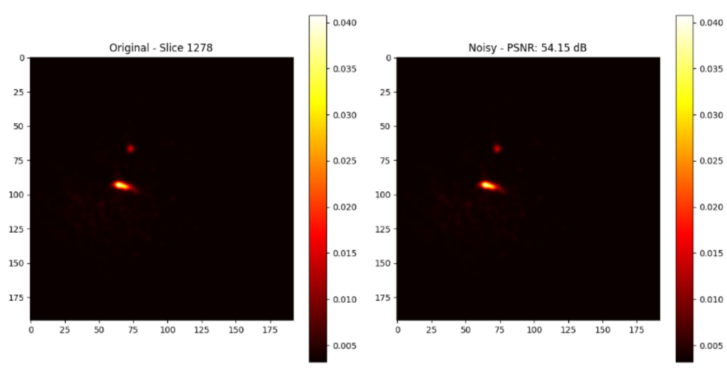
Motion Parameters:

Probability (20%): A realistic occurrence rate for slice-to-slice movement

Max Shift (2 pixels): Equivalent to roughly 2-4mm displacement

Direction: Random x,y shifts imitating head or body movement

Pattern: Intermittent, impacting non-consecutive slices



Original Image Image with Motion Artifacts

Fig 3.3: Slices view under Motion Artifacts Noise

CHAPTER-4

METHODOLOGY

4.1 Introduction

This chapter outlines a thorough methodology for reducing noise in dynamic Positron Emission Tomography (PET) images using Deep Autoencoders (DAEs). The main aim is to improve image quality by effectively diminishing noise while retaining vital anatomical and functional details that are crucial for precise clinical diagnoses. PET imaging, especially in situations involving low-dose acquisitions, is prone to inherent noise due to the limited number of detected photons and the random nature of radioactive decay. This approach tackles these issues with advanced deep learning techniques specifically developed for applications in medical imaging.

The method employs a 3D patch-based Deep Autoencoder framework that handles volumetric PET data using dimensionality reduction and reconstruction methods. In contrast to conventional denoising approaches that depend on manually designed filters or statistical hypotheses, the suggested DAE extracts optimal noise removal patterns directly from the data, facilitating enhanced performance in maintaining both structural fidelity and quantitative precision of PET images.

The framework includes several essential elements: a thorough review of existing denoising strategies, a detailed rationale for the choice of DAE, a comprehensive architectural design, robust training methodologies, and strict evaluation standards. Each element is crafted to tackle specific obstacles in denoising PET images, such as managing 3D volumetric data, maintaining tracer uptake patterns, and ensuring computational efficiency for use in clinical settings.

Denoising medical images is a critical preprocessing step that has a direct effect on diagnostic precision and clinical decision-making processes. The advancement of denoising techniques has shifted from basic linear filters to more sophisticated deep learning frameworks, each presenting unique benefits and drawbacks within the realm of PET imaging.

4.2 DICOM PET Data:

Positron Emission Tomography (PET) imaging data is typically stored in the DICOM (Digital Imaging and

Communications in Medicine) format, which serves as the standard for managing, storing, and transmitting medical images. Each PET scan is composed of multiple two-dimensional (2D) image slices, with every slice acting as a cross-sectional view of the patient's anatomy at a particular spatial location. These 2D slices are saved as separate DICOM files that include matrix-structured pixel intensity values arranged in rows and columns. The pixel values indicate the distribution of a radioactive tracer within the body, which reflects the metabolic or functional activity of various tissues. This functional imaging plays a crucial role in the diagnosis and monitoring of a range of conditions, including cancer, neurological disorders, and cardiovascular diseases. In addition to pixel data, DICOM files contain critical metadata, such as patient details, acquisition settings, and image orientation, all of which are essential for accurate image reconstruction and analysis.

Dynamic PET imaging captures multiple image frames over time, allowing for the observation of changes in tracer concentration within the body. Each 2D DICOM file therefore, provides not only spatial information but also temporal data, representing tracer uptake at a specific moment. However, individual slices alone do not suffice for thorough analysis, as clinical interpretation and computational processing necessitate the integration of these slices into cohesive three-dimensional (3D) volumes while accounting for their temporal changes. Adequate parsing and comprehension of each DICOM file, incorporating both pixel data and metadata, are crucial to ensure spatial and temporal consistency during the reconstruction of volumes. This foundational procedure guarantees the accurate alignment of slices and time intervals, which is vital for subsequent operations such as patch extraction, model training, and eventually, the denoising of PET images through deep learning techniques.

The dataset utilized in this research comprises 4D DICOM chest PET scan images acquired from Turun Yliopistollinen sairaala (Turku University Hospital) in Turku, Finland, using a GE Medical Systems Discovery MI PET scanner (station name: "Aino"). This dataset comprises 4D dynamic PET scans divided into 24 sub-datasets, each containing approximately 1,704 images obtained with different radiotracer doses (e.g., 20%, 40%) and energy windows (e.g., 150–120 eV), and appropriately labeled. The images have been reconstructed using two widely used algorithms: OSEM, known for its rapid convergence, and BSREM, which enhances noise reduction while maintaining detail. Every PET slice features a resolution of 192×192 pixels, with a slice thickness of 3.75 mm. This realistic resolution, combined with variability in tracer activity and reconstruction techniques, offers a rich dataset encompassing a range of noise patterns, making it ideal for training Deep Autoencoder models aimed at effectively denoising PET images, particularly in simulating low-dose scenarios to minimize radiation exposure while preserving image quality.

The project starts with the processing of PET data in DICOM (Digital Imaging and Communications in Medicine) format. Every DICOM file corresponds to an individual 2D slice of the PET acquisition, containing pixel data along with detailed metadata that includes patient details, acquisition parameters, and geometric positioning information. The DICOM files are organized hierarchically, with various series representing differing acquisition conditions, time points, or reconstruction settings. Each file holds key metadata fields such as:

- SeriesInstanceUID: Unique identifier for clustering related slices
- SliceLocation: Spatial positioning information for volumetric reconstruction
- InstanceNumber: Information related to the sequence of slices
- RescaleSlope and RescaleIntercept: Calibration parameters for quantitative analysis
- Rows and Columns: Dimensions of the image
- Modality: Confirmation of the PET imaging type

The DICOM loading procedure features a strong system for file detection and verification.

```python
def load_dicom_files(directory):
    """Load all DICOM files from a directory structure."""
    dicom_files = []

    for root, _, files in os.walk(directory):
        for file in files:
            if file.lower().endswith('.dcm'):
                dicom_files.append(os.path.join(root, file))

    return dicom_files
```

This recursive method guarantees a thorough exploration of DICOM files, no matter how complicated the directory structure may be. The algorithm checks the format and accessibility of each file prior to adding it to the processing pipeline.

4.3 Series Organization:

The process of organizing series is a vital step in managing PET DICOM data, where separate 2D slices are assembled into coherent series based on the unique identifier known as the SeriesInstanceUID. Each SeriesInstanceUID is linked to a specific acquisition session or protocol, enabling the system to group slices that originate from the same scanning sequence. This categorization guarantees that images captured under the same conditions, such as spatial positioning and acquisition timing, are assembled together.

By arranging slices into series, it becomes feasible to create accurate 3D volumes that reflect the patient's anatomy and tracer distribution during particular time intervals. This phase is crucial since PET scans typically comprise multiple series tied to various scanning phases, acquisition protocols, or imaging sequences. Without this organization, mixing slices from diverse sessions might result in inaccuracies in volume reconstruction and undermine the precision of subsequent tasks like image denoising or quantitative analysis.

Additionally, organizing by series helps maintain the distinction between acquisitions with different parameters, such as dose, energy window, or reconstruction algorithm. This distinction is essential for the integrity and consistency of data, particularly when developing machine learning models. Processing slices categorized by SeriesInstanceUID preserves the clinical and temporal context of each scan, ensuring that the resulting 3D volumes accurately reflect the physiological and pathological information crucial for effective diagnosis and treatment planning.

```python
def organize_pet_series(dicom_files):
    """Organize DICOM files into different PET series."""
    series_dict = {}

    for file_path in dicom_files:
        ds = pydicom.dcmread(file_path)

        if hasattr(ds, 'Modality') and ds.Modality != 'PT':
            continue

        series_id = ds.SeriesInstanceUID

        if series_id not in series_dict:
            series_dict[series_id] = []

        series_dict[series_id].append((file_path, ds))

    return series_dict
```

4.4 Volume Building and Normalization:

Once the individual 2D DICOM slices have been organized into coherent series, the subsequent step involves reconstructing these slices into 3D volumes that depict the spatial structure of the scanned area. The process of volume construction entails layering the ordered slices along the axial axis according to their spatial metadata, including slice positioning and orientation, to create a continuous three-dimensional visualization of the patient's anatomy. This volumetric model is essential for maintaining spatial relationships between slices and enabling further 3D analysis and processing.

Alongside spatial reconstruction, intensity normalization is applied during the volume construction phase to equalize pixel values across various PET acquisitions. Given that PET scans can exhibit significant variability due to differences in scanner calibration, tracer administration, and acquisition methods, normalization ensures that pixel intensities remain within a consistent range, which aids downstream algorithms in processing uniformly. A popular method employed is min-max normalization, which linearly scales all pixel values to the range [0, 1]. This standardization improves the stability and convergence of deep learning models by mitigating variations resulting from different acquisition conditions, thereby enhancing overall model performance.

The combination of volume reconstruction and intensity normalization yields a standardized 3D input that accurately captures both the spatial and intensity attributes of the original PET scan. This processed volume acts as the basis for subsequent processing steps, including patch extraction, noise simulation, and model training, which enable reliable and effective denoising of dynamic PET images utilizing deep autoencoder architectures.

```python
def normalize_volume(volume):
    """Normalize volume to 0-1 range."""
    min_val = np.min(volume)
    max_val = np.max(volume)

    if max_val > min_val:
        return (volume - min_val) / (max_val - min_val)
    else:
        return volume
```

Advantages of Normalization:

- The normalization approach offers multiple benefits:
- Model convergence: Enhanced stability during training and faster convergence times.
- Gradient flow: Improved gradient transmission throughout deep networks.
- Cross-patient consistency: Unified intensity ranges among different individuals.
- Activation function optimization: Maximized use of sigmoid/tanh function ranges.

4.5 Adding Noise to PET Images for Training:

To develop a supervised denoising model, it is essential to have paired datasets containing both noisy and clean images. Since it is often difficult to find actual noisy-clean pairs for PET imaging, synthetic noise is applied to clean images to mimic realistic degradation. The typical noise types incorporated include:

1. Poisson Noise:

Also referred to as shot noise, Poisson noise is a natural occurrence resulting from photon counting statistics in PET imaging. The count of detected photons fluctuates randomly following a Poisson distribution, leading to noise that is dependent on the signal—regions with greater tracer uptake usually have higher photon counts and, consequently, lower noise, while areas with less activity tend to exhibit higher noise levels. This type of noise is intrinsic to the PET acquisition process and is vital for accurately simulating scan degradation.

```python
def add_poisson_noise(clean_data, snr=10):
    """Add Poisson noise to simulate low-dose PET acquisition."""
    # Scale the data
    scaled_data = clean_data * snr

    # Add Poisson noise
    noisy_data = np.random.poisson(scaled_data)

    # Scale back to original range
    noisy_data = noisy_data / snr

    # Clip to [0, 1] for normalized data
    noisy_data = np.clip(noisy_data, 0, 1)

    return noisy_data
```

2. Gaussian Noise:

Gaussian noise represents the electronic and system noise generated by the PET scanner hardware. This noise is unrelated to signal intensity and is typically described by a normal distribution with a defined mean and variance. Gaussian noise simulates issues such as electronic variability and thermal noise in the detectors and image reconstruction methods, which uniformly reduce image quality throughout the scan.

```python
def add_gaussian_noise(clean_data, sigma=0.05):
    """Add Gaussian noise to simulate electronic noise."""
    noise = np.random.normal(0, sigma, clean_data.shape)
    noisy_data = clean_data + noise
    return np.clip(noisy_data, 0, 1)
```

3. Motion Artifacts:

Motion artifacts result from patient movement during the PET scan, which can occur because of breathing, involuntary muscle actions, or discomfort while being scanned. These artifacts appear as blurring, streaking, or ghosting in the reconstructed images, which diminishes anatomical accuracy. To simulate motion artifacts, distortions or transformations are applied that replicate real patient movements, assisting the model in developing resilience against these frequent clinical challenges.

By artificially introducing these types of noise to clean PET images, the training dataset accurately represents the difficulties faced in clinical PET imaging. This allows the deep autoencoder model to effectively understand noise patterns and improve denoising capabilities during inference.

4.6 3D Patch Extraction

The methodology utilizes a patch-based technique to address the computational demands of 3D PET volumes while preserving spatial context. This approach segments large volumes into smaller, more manageable 3D patches that can be processed efficiently by the deep learning model.

In this method, patch-based processing involves meticulous optimization of patch dimensions, stride, and spatial tracking to achieve effective and precise denoising. A patch size of 5×5×5 presents a practical balance between memory efficiency, adequate spatial context, and effective feature extraction, making it suitable for identifying noise patterns while facilitating rapid processing. A stride of 2 creates overlapping patches, which improves output quality by minimizing boundary artifacts, fostering smoother transitions, averaging noise impacts, and covering missing details through redundancy. Accurate position tracking for each patch is ensured using 3D coordinates, which maintains proper placement during volume reconstruction, manages overlapping areas effectively, and guarantees complete coverage of the input volume, including special attention to boundary regions.

```
def extract_patches(volume, patch_size=(5, 5, 5), stride=2):
    """Extract 3D patches from a volume."""
    Z, X, Y = volume.shape
    patches = []
    positions = []

    for z in range(0, Z - patch_size[0] + 1, stride):
        for x in range(0, X - patch_size[1] + 1, stride):
            for y in range(0, Y - patch_size[2] + 1, stride):
                patch = volume[z:z+patch_size[0],
                               x:x+patch_size[1],
                               y:y+patch_size[2]]

                patch_vector = patch.flatten()
                patches.append(patch_vector)
                positions.append((z, x, y))

    return np.array(patches), positions
```

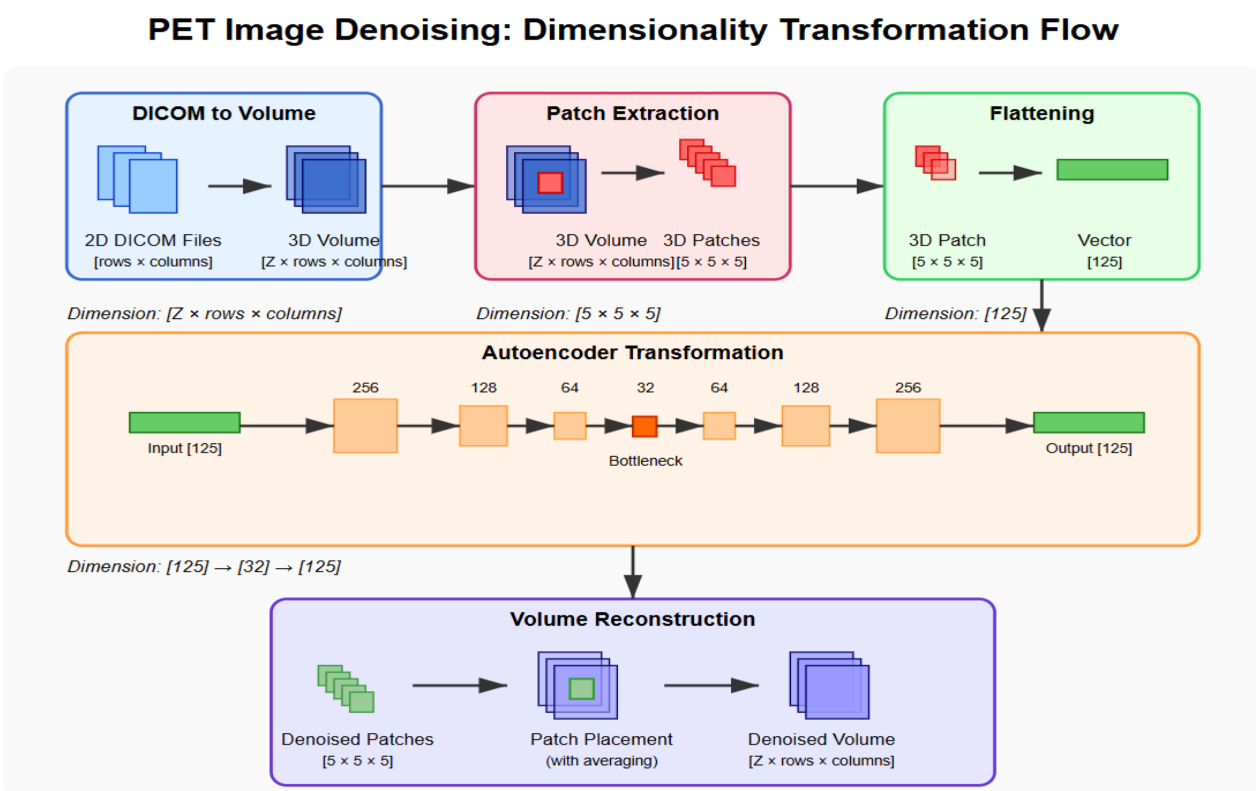


Fig 4.1 Patch Extraction

4.7 Deep Autoencoder Architecture and Training:

The structure and training of the deep autoencoder designed for denoising PET images are specifically crafted to meet the distinct challenges associated with the noisy, high-dimensional characteristics of PET data. The framework is developed to learn effective feature representations that encapsulate the critical structures of clean PET images while simultaneously filtering out noise. This section outlines the architectural elements of the autoencoder and the detailed training approach utilized to enhance model efficacy.

Autoencoder Framework Specification

The autoencoder processes 3D patches taken from the normalized PET volumes. Each input patch measures 5×5×5 voxels in spatial dimensions, which are transformed into a one-dimensional vector of 125 in length for input into the network. This flattening technique facilitates the use of fully connected (dense) layers, which are particularly adept at identifying global patterns within these relatively compact patches. This method differs from convolutional architectures by emphasizing compact feature representations through a fully connected structure.

The encoder pathway is composed of a series of dense layers with decreasing numbers of neurons: 256, 128, 64, and ultimately, 32 neurons at the bottleneck. This exponential reduction compels the network to compress the input data, driving the identification of compact and noise-resistant representations of the original image content. Each dense layer employs the Exponential Linear Unit (ELU) activation function. The ELU function provides advantages over more conventional activations, such as ReLU, by producing smooth gradients even for negative inputs, thus preventing "dead neurons" and facilitating more stable and efficient training. Following each dense layer, batch normalization is implemented to standardize activations to a zero mean and unit variance within each mini-batch. This normalization reduces internal covariate shift, expedites convergence, and enhances overall model robustness.

To avoid overfitting, dropout regularization is applied after batch normalization with a relatively low dropout rate of 0.1. This low rate strikes a balance between regularization and maintaining the subtle details essential for precise denoising. Excessive dropout could result in the loss of critical information, particularly in medical imaging, where it is vital to preserve fine anatomical details. The bottleneck layer, consisting of 32 neurons, serves as the fundamental compressed representation, pushing the network to encode the most prominent features of the clean PET patches while eliminating noise.

Reflecting the encoder structure, the decoder pathway gradually enlarges the compressed bottleneck representation back to the original input dimensions. It utilizes dense layers sized 64, 128, 256, and ultimately outputs a 125-dimensional vector that corresponds to the reconstructed patch. The symmetric design preserves architectural consistency, ensuring that the decoder possesses adequate capacity to accurately reconstruct detailed image content from the compressed features. The output layer employs a sigmoid activation function that confines pixel intensity predictions to the normalized range of [0,1]. This restriction aligns with the preprocessing normalization, guaranteeing a consistent data representation throughout the entire pipeline.

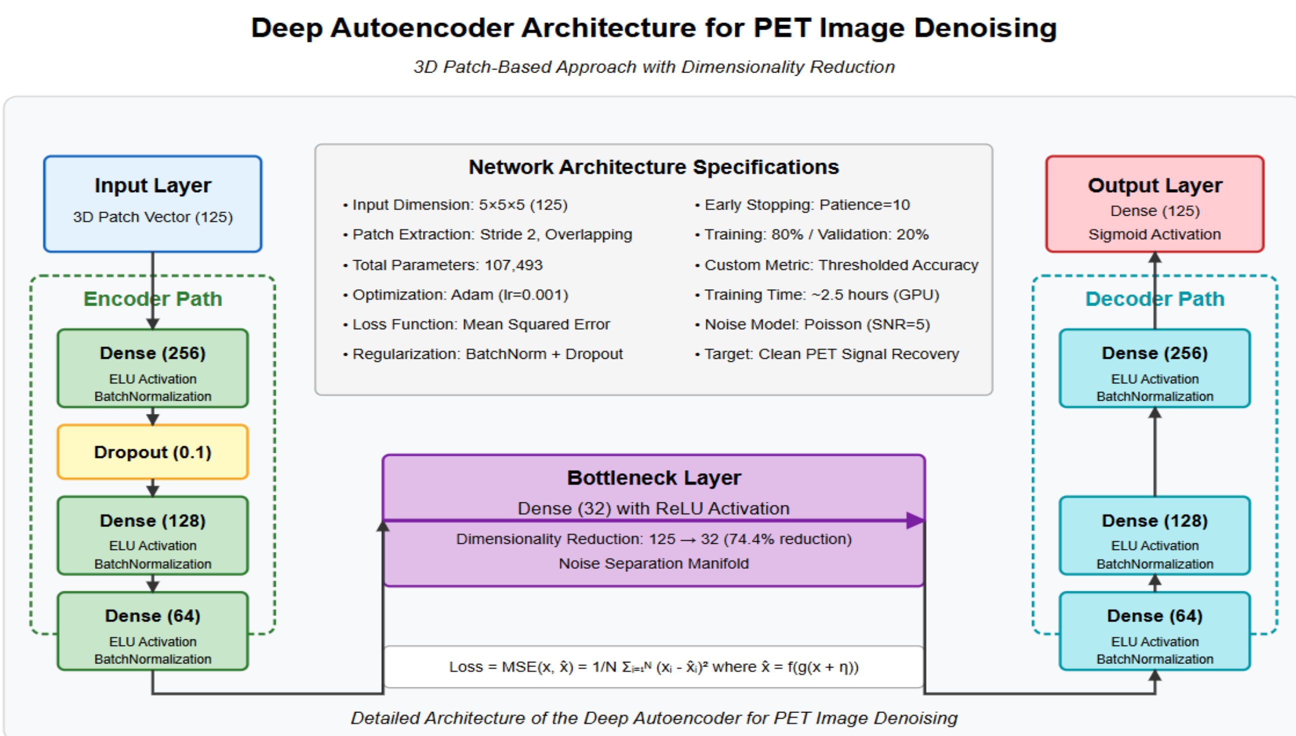


Figure 4.2: DAE Architecture

Overall Architecture Philosophy

The architecture adopts a symmetric encoder-decoder framework with specific adjustments tailored for medical imaging tasks. The underlying design philosophy underscores three primary principles: preserving information through regulated dimensionality reduction, learning features hierarchically to effectively reduce multi-scale noise, and achieving computational efficiency for use in clinical settings. The network analyzes 3D patches derived from PET volumes, allowing for the simultaneous assessment of spatial relationships across all three dimensions. This method maintains anatomical continuity that could be compromised in slice-by-slice processing, while still being computationally manageable compared to processing entire volumes.

Input Layer and Patch Processing

3D Patch Extraction Strategy: The input layer takes 3D patches that are 5×5×5 voxels in size, which are then transformed into 125-dimensional vectors. This patch size strikes an ideal balance between adequately capturing spatial context and ensuring computational efficiency. If the patches were smaller, they might overlook significant anatomical connections, whereas larger patches would substantially heighten computational demands and memory usage.

```python
# Patch extraction with overlapping strategy
def extract_patches(volume, patch_size=(5, 5, 5), stride=2):
    patches = []
    positions = []
    Z, X, Y = volume.shape

    for z in range(0, Z - patch_size[0] + 1, stride):
        for x in range(0, X - patch_size[1] + 1, stride):
            for y in range(0, Y - patch_size[2] + 1, stride):
                patch = volume[z:z+patch_size[0],
                              x:x+patch_size[1],
                              y:y+patch_size[2]]
                patches.append(patch.flatten())
                positions.append((z, x, y))

    return np.array(patches), positions
```

Overlapping Patch Strategy: The extraction process with a stride of 2 produces overlapping patches that facilitate smooth reconstruction and minimize boundary artifacts. During the reconstruction phase, the overlapping sections are averaged, yielding a natural smoothing effect that mitigates possible checkerboard patterns or discontinuities. This overlapping approach is essential for ensuring spatial consistency in the denoised volume.

Normalization and Pre-processing: Input patches are subject to volume-wise normalization within the [0,1] range, which guarantees stable training dynamics and uniform performance across various patient studies. This normalization step maintains the relative intensity relationships within each volume while standardizing the dynamic range of inputs for the network.

Encoder Architecture:

The encoder component achieves progressive dimensionality reduction by utilizing a series of fully connected layers with diminishing numbers of neurons. This structure fosters a hierarchical abstraction of input features, with each layer capturing increasingly intricate and abstract representations.

Layer-by-Layer Analysis:

First Encoding Layer (125 → 256):

```
inputs = Input(shape=(input_dim,))  # input_dim = 125
x = Dense(256, activation='elu')(inputs)
x = BatchNormalization()(x)
x = Dropout(0.1)(x)
```

The first step of expanding from 125 to 256 dimensions enables the network to investigate a more diverse representation space prior to starting compression. The ELU (Exponential Linear Unit) activation function offers smooth gradients for negative inputs and keeps computational efficiency intact. Batch normalization enhances training stability by normalizing the inputs of layers, and a dropout rate of 10% helps to mitigate overfitting during the training process.

Second Encoding Layer (256 → 128):

```
x = Dense(128, activation='elu')(x)
x = BatchNormalization()(x)
```

The initial compression phase decreases the dimensionality by half, compelling the network to recognize and preserve the most significant features while eliminating unnecessary information. This layer initiates the essential process of differentiating between noise and signal by determining which input elements add meaningful content to the image as opposed to noise.

Third Encoding Layer (128 → 64):

```
x = Dense(64, activation='elu')(x)
x = BatchNormalization()(x)
```

A further reduction to 64 dimensions results in a 74.4% decrease from the original size of the input. At this stage, the network acquires highly abstract representations that encapsulate key anatomical and functional patterns while diminishing noise elements. The representations at this stage ought to be resilient to variations in noise while maintaining important image information.

Bottleneck Layer Design

Critical Dimensionality Reduction:

```
x = Dense(32, activation='relu')(x)  # Bottleneck layer
```

The bottleneck layer presents the most condensed representation with just 32 dimensions, achieving a 74.4% decrease from the preceding layer and an overall compression of 74.4%. This significant compression compels the network to identify only the most crucial features necessary for image reconstruction. The bottleneck functions as a noise separation manifold, where signal elements are maintained while noise elements are eliminated.

Regarding Activation Function Selection:

The bottleneck opts for ReLU activation rather than ELU, resulting in distinct mathematical behavior that fosters a sparse representation. The hard thresholding nature of ReLU at zero inherently dampens weak features likely linked to noise, while it keeps robust features tied to anatomical structures and tracer uptake.

Information Bottleneck Theory:

The design of the bottleneck adheres to information bottleneck principles, where the compressed representation retains minimal information needed for precise reconstruction. This compression inherently filters out noise elements that do not add value to meaningful image content, while maintaining crucial information for anatomical and functional accuracy.

Decoder Architecture

The decoder replicates the structure of the encoder but in reverse order, gradually returning the compressed representation to its original input dimensions. Each layer of the decoder learns to rebuild the image's finer details using the abstract features extracted from the bottleneck.

Progressive Reconstruction:

First Decoder Layer (32 → 64):

```
x = Dense(64, activation='elu')(x)
x = BatchNormalization()(x)
```

The first phase of expansion starts by rebuilding spatial patterns from the reduced representation. This layer acquires the skill to translate abstract features into more tangible spatial patterns that will direct the next stages of reconstruction.

Second Decoder Layer (64 → 128):

```
x = Dense(128, activation='elu')(x)
x = BatchNormalization()(x)
```

Ongoing expansion enhances spatial detail and starts to recreate intricate anatomical characteristics. The network acquires the ability to produce detailed elements of the image using the essential features retained from the bottleneck.

Third Decoder Layer (128 → 256):

```
x = Dense(256, activation='elu')(x)
x = BatchNormalization()(x)
```

The last stage of expansion gets ready for output reconstruction by creating detailed feature representations that will be transformed back into pixel values. This layer maintains a balance between reconstructing details and suppressing noise that was learned throughout the training process.

Output Layer (256 → 125):

```
outputs = Dense(125, activation='sigmoid')(x)
```

The output layer rebuilds the original 125-dimensional patch vector by applying a sigmoid activation function, which restricts the outputs to a range of [0,1], in alignment with the normalized input data. This layer captures the ultimate transformation from abstract features to clear pixel values.

Architectural Strategies:

Revised Batch Normalization Integration: Batch normalization is utilized after every hidden layer to enhance the stability of training and expedite convergence. In the context of medical imaging, batch normalization ensures consistent feature scales across various patient studies and imaging protocols, which enhances generalization performance.

Revised Dropout Strategy: A targeted dropout rate of 10% is implemented solely in the early encoder layer to mitigate overfitting while allowing information to flow through the bottleneck. This minimal level of dropout retains crucial information for accurate reconstruction while offering adequate regularization for effective training.

Revised Activation Function Selection: The deliberate selection of ELU and ReLU activation functions is made: ELU delivers smooth gradients throughout the majority of the network, whereas ReLU in the bottleneck facilitates natural sparsity that helps in reducing noise. The use of sigmoid activation in the output layer ensures that the output remains within the appropriate range, aligning with the normalized input data.

Training Strategy and Optimization:

The strategy for training the deep autoencoder is meticulously crafted to ensure strong denoising performance while avoiding issues related to overfitting and convergence. The model utilizes paired datasets consisting of noisy and clean 3D patches, allowing the network to learn to associate noisy inputs with their clean versions. The training process aims to optimize pixel-wise reconstruction accuracy while incorporating strategies to maintain generalization capabilities.

The Adam optimizer is chosen for its ability to adapt the learning rate and for its quick convergence properties. An initial learning rate of 0.001 is set, supplemented by an adaptive scheduling system that modifies the learning rate dynamically in response to training development. This versatility enables the network to efficiently navigate complex loss landscapes, avoiding poor local minima and speeding up the training process. Adam's robustness across a range of deep learning tasks also simplifies hyperparameter adjustments.

The main loss function employed is the mean squared error (MSE) between the predicted denoised patch and the actual clean patch. MSE penalizes differences in pixel intensities, motivating the network to evenly minimize reconstruction error across the patch. Such pixel-level accuracy is essential for medical imaging, where minor intensity differences can signify important physiological changes.

To provide clear feedback on training progress, a custom metric named thresholded accuracy is calculated during both training and validation. This metric deems a prediction accurate if the intensity of the reconstructed pixel deviates from the ground truth by less than a predetermined tolerance (0.1 in this scenario). This metric serves as a complement to MSE by measuring the frequency with which the model achieves acceptable reconstruction quality, facilitating a qualitative evaluation of training advancement.

To mitigate overfitting and enhance model generalization, early stopping is implemented with a patience setting of 10 epochs, tracking validation loss. Training ceases if there is no improvement in validation performance over 10 consecutive epochs, ensuring that the best weights are retained before any decline occurs. This method not only saves computational resources but also minimizes the likelihood of overfitting to training noise patterns.

Additionally, model checkpointing is employed to preserve the best-performing model weights based on validation loss throughout the training process. This guarantees that the final model deployed corresponds to the peak training performance, even if subsequent epochs lead to a decline. The dataset is divided, reserving 20% for validation, which provides a solid assessment of generalization effectiveness during training.

Collectively, these architectural decisions and training approaches create a comprehensive framework that allows the deep autoencoder to effectively learn noise reduction while retaining crucial details in PET images. The balance struck between compression and reconstruction capability, along with contemporary optimization and regularization strategies, ensures efficient, stable, and reproducible denoising tailored for dynamic PET imaging applications.

4.8 PET Image Denoising Pipeline:

The PET image denoising process is a systematic, comprehensive procedure aimed at improving the quality of noisy PET scans through a patch-based deep autoencoder methodology. Initially, the raw medical imaging data, generally saved as 2D DICOM files, is structured into a unified 3D volume. Each DICOM file corresponds to an individual image slice, and by aligning them with metadata such as SeriesInstanceUID, a complete volumetric representation of the PET scan is formed. This 3D volume encompasses the spatial anatomy essential for detailed processing and analysis.

After constructing the 3D volume, it is divided into smaller spatial sections known as patches. Each patch, sized 5×5×5 voxels, captures localized imaging features. These patches are then converted into 125-

dimensional vectors to be utilized as input for the deep autoencoder model. This patch-based strategy enhances computational efficiency and allows the model to target localized noise patterns for denoising while retaining anatomical details. It also guarantees that the network manages high-dimensional medical data in more digestible segments, improving both learning and generalization.

Central to the pipeline is the deep autoencoder neural network. It compresses each 125-dimensional patch through a sequence of dense layers—successively diminishing dimensions from 256 to 32 neurons—until it reaches a bottleneck, which captures the most critical features of the clean PET signal. The decoder then reconstructs the information by reversing the compression process, enlarging the compact representation back to its original dimensions. Following the denoising of each patch, the output vectors are reshaped into their 3D format and repositioned within their original spatial arrangement. Overlapping patches are averaged to ensure smooth transitions and continuity at the boundaries. The end result is a completely denoised 3D PET image that preserves both anatomical and functional integrity while significantly minimizing noise artifacts.

CHAPTER-5
IMPLEMENTATION AND RESULTS

5.1 Dataset Overview

The dataset utilized in this research comprises 4D DICOM chest PET scan images acquired from Turun Yliopistollinen sairaala (Turku University Hospital) in Turku, Finland, using a GE Medical Systems Discovery MI PET scanner (station name: "Aino"). This extensive dataset facilitates dynamic imaging analysis and is particularly ideal for studies focused on PET image denoising due to its high-quality annotations and uniform acquisition protocols. The dataset is systematically organized into 24 sub-datasets, each containing around 1,740 images, ensuring a diverse range of data samples. These sub-datasets are categorized based on parameters like tracer activity and reconstruction method. For example, naming patterns such as PT_20p 150_120 OSEM and PT_20p 150_120 BSREM indicate different percentages of tracer activity (20% and 40%) and distinct energy windows (150–120 eV or 120–150 eV) during the image acquisition.

The figures 20p and 40p represent the percentage of the administered radiotracer, enabling the emulation of low-dose acquisition settings for developing and evaluating denoising algorithms. The designations 150_120 and 120_150 denote the range of photon energies, expressed in electron volts (eV), emitted by the tracer, which provides different spectral attributes to examine noise behavior under varying physical conditions. The images are reconstructed utilizing two main algorithms: BSREM (Block Sequential Regularized Expectation Maximization) and OSEM (Ordered Subset Expectation Maximization). These methods are commonly employed in both clinical and research PET imaging. BSREM is particularly recognized for its excellent noise reduction and edge-preserving qualities, while OSEM is a conventional iterative approach that offers quicker convergence and is widely utilized in commercial PET/CT systems.

Each PET slice features a matrix size of 192×192 pixels, with the quantity of slices per volume potentially varying. The thickness of each slice is close to 3.75 mm, which is a standard configuration that balances anatomical detail with data volume. These specifications are conducive to training deep learning models, particularly denoising autoencoders, in realistic and clinically applicable conditions.

5.2 Tools and Libraries

This experimentation and implementation occurred on Google Colab, a powerful cloud platform that provides free access to GPU resources, significantly accelerating the training for intricate deep neural network models.

1. **TensorFlow/Keras** serves as the main deep learning framework utilized for constructing and training the denoising autoencoder model. The code makes use of Keras' high-level API for building neural network layers, defining the model structure, and training with features like early stopping and model checkpoints. It establishes the basis for the deep autoencoder that learns to convert noisy PET images back to their clean counterparts.

2. **NumPy** acts as the essential numerical computing library, managing all array operations and mathematical tasks. It is heavily employed for volume manipulation, patch extraction, normalization, noise generation, and metric evaluations like MSE and PSNR. Throughout the processing pipeline, NumPy arrays serve as the main data structure for storing the 3D PET volumes.

3. **Nibabel** is a dedicated neuroimaging library that facilitates reading and writing neuroimaging file formats. Although imported for this project, it appears to be intended for working with NIfTI format files, while the current implementation focuses on DICOM files. It would typically be used to load brain imaging data in standardized formats.

4. **PyDICOM** is the critical library for reading and interpreting DICOM (Digital Imaging and Communications in Medicine) files, which is the standard format for medical imaging data. It manages the complex extraction of DICOM metadata, including series organization, slice positioning, and pixel data rescaling, enabling the transformation of raw DICOM files into usable 3D numpy arrays.

5. **Matplotlib** is extensively utilized for visualization and saving output images. It generates comparison plots that illustrate original, noisy, and denoised PET images from various viewing planes (axial, coronal, sagittal), produces training history plots, devises maximum intensity projections (MIP), and creates difference analysis visualizations to showcase the effectiveness of denoising.

6. **SciPy** offers advanced scientific computing capabilities, with the scipy.ndimage.gaussian_filter specifically imported for potential image filtering tasks. While it is not heavily utilized in the current application, it provides additional image processing functionalities that could be incorporated into preprocessing or post-processing phases.

7. **Google Colab** Integration includes specific imports (google.colab.drive) for mounting Google Drive, facilitating cloud-based execution with access to stored DICOM files and the ability to save results. This arrangement allows for processing extensive medical imaging datasets without the limitations of local storage.

8. **tqdm** is a library for progress bars that gives visual feedback during lengthy operations such as reading multiple DICOM files, constructing 3D volumes from image stacks, and processing patches. It enhances the user experience by displaying real-time progress during extensive medical image processing tasks.

5.3 Evaluation Metrics

MSE (Mean Square Error): Mean Square Error is a fundamental yet extensively utilized metric in image processing that quantifies the average squared differences between the pixel values of the original (ground truth) and the processed (denoised) images. It provides a straightforward numerical indication of reconstruction accuracy. A smaller MSE value suggests that the denoised image more closely resembles the original image, indicating higher quality. Nevertheless, MSE does not consider human visual perception and may not accurately represent perceptual quality. It is calculated as follows:

$$\mathrm{MSE} = \frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} [I(i,j) - K(i,j)]^2$$

In the equation, I(i, j) represents the pixel intensity of the original image, while K(i, j) denotes the corresponding pixel intensity of the denoised image, with m and n indicating the dimensions of the image.

PSNR (Peak Signal-to-Noise Ratio): It is based on MSE and indicates the ratio between the maximum possible pixel value of an image and the intensity of the noise that impacts its quality. Typically expressed in decibels (dB), it serves as a standardized method for comparing image quality. Higher PSNR values indicate improved image fidelity, with values exceeding 30 dB generally regarded as satisfactory for most medical imaging applications. Although PSNR is a straightforward and easily understandable metric, like MSE, it may not always correlate well with perceived image quality. The formula is:

$$\mathrm{PSNR} = 10 \cdot \log_{10} \left(\frac{\mathrm{MAX}^2}{\mathrm{MSE}} \right) \ \mathrm{dB}$$

where MAX is the maximum possible pixel value of the image (often 1 for normalized images or 255 for 8-bit images).

SSIM (Structural Similarity Index Measure): SSIM is a more sophisticated perceptual metric that assesses image quality based on structural details, brightness, and contrast, aligning it more closely with human visual perception. Unlike MSE and PSNR, SSIM takes into account the spatial relationships of pixels and the preservation of structural patterns between the original and processed images. SSIM scores range from -1 to 1, although values below 0 are very uncommon in practice; a score nearer to 1 signifies greater structural similarity. It is especially valuable in medical imaging, where maintaining anatomical structures is essential. The formula for SSIM is:

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

where μ and σ represent the mean and variance of the image patches, and C_1, C_2 are constants to stabilize the division.

BRISQUE (Blind/Referenceless Image Spatial Quality Evaluator): BRISQUE (Blind/Referenceless Image Spatial Quality Evaluator) is a metric for image quality assessment that operates without the need for a reference image, assessing the naturalness and perceived quality of an image. It employs machine learning techniques, specifically Support Vector Machines, which are trained on Natural Scene Statistics (NSS) to identify distortions by examining local structures within the image. Unlike PSNR and SSIM, both of which require a reference image for comparison, BRISQUE is particularly advantageous in practical situations where a reference may not exist. A lower BRISQUE score signifies a better perceived image quality. Although it lacks a straightforward closed-form expression similar to MSE or PSNR, it is determined by analyzing statistical features gathered from spatial domain patches of the image and using a trained regression model for computation. Its use is on the rise for subjective image quality evaluation across various fields, including medical imaging.

5.4 Results of Different Noise Types

Slices View Under Poisson Noise

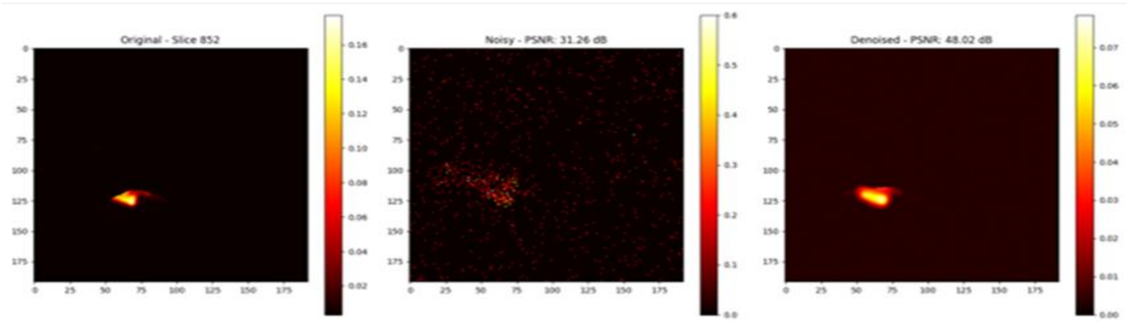


Figure 5.1 Original Slice 0 Noisy Slice Denoised Slice

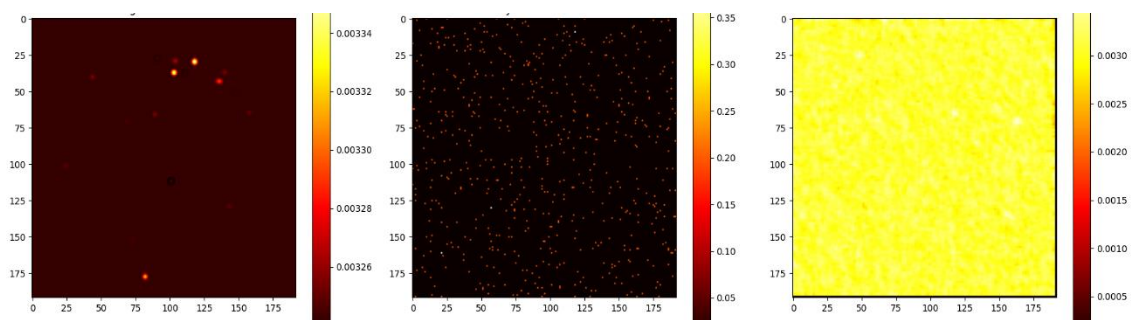


Figure 5.2 Original Slice 1 Noisy Slice Denoised Slice

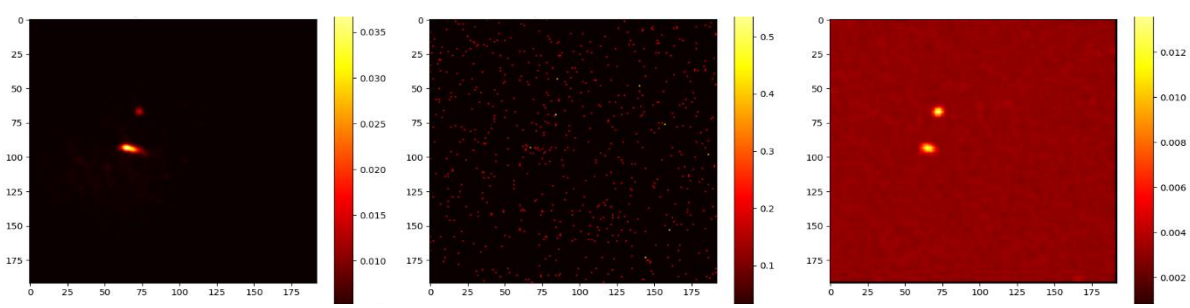


Figure 5.3 Original Slice 2 Noisy Slice Denoised Slice

ANATOMICAL PLANE VIEWS (POISSON NOISE):

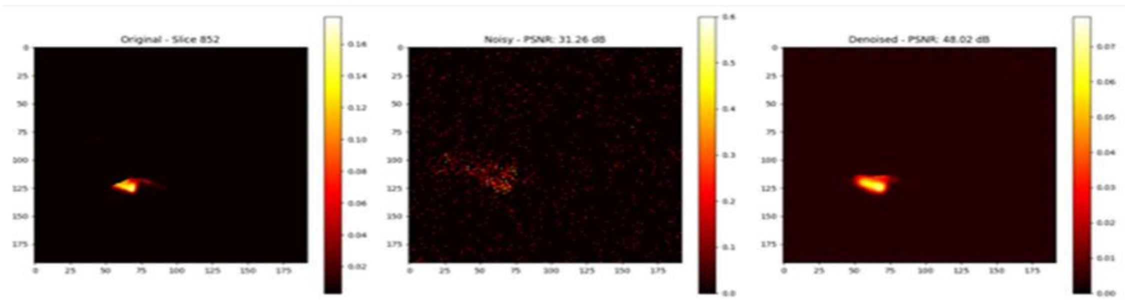


Figure 5.4 AXIAL VIEW COMPARISON

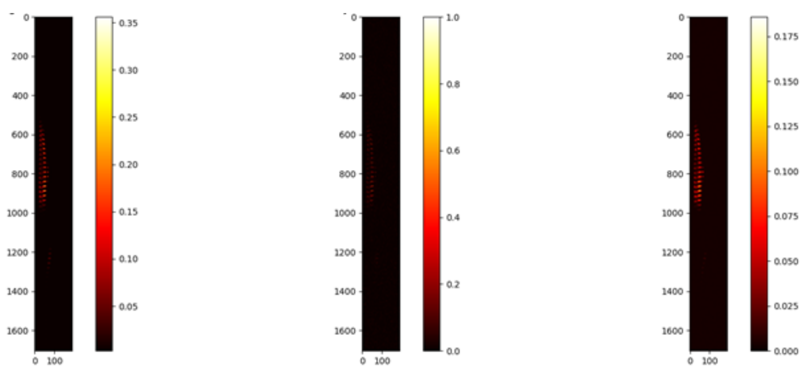


Figure 5.5 CORONAL VIEW COMPARISON

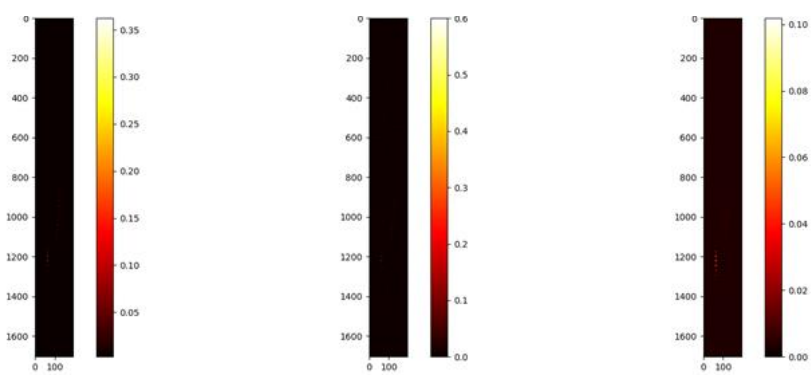


Figure 5.6 SAGITTAL VIEW COMPARISON

Slices View Under Gaussian Noise

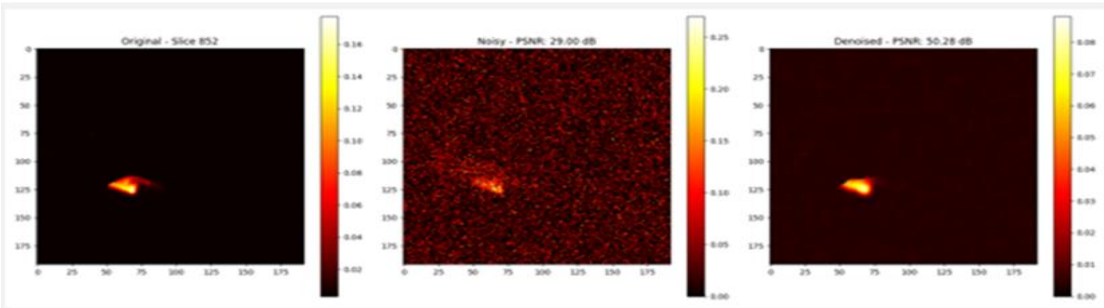


Figure 5.7 Original Slice 0 Noisy Slice Denoised Slice

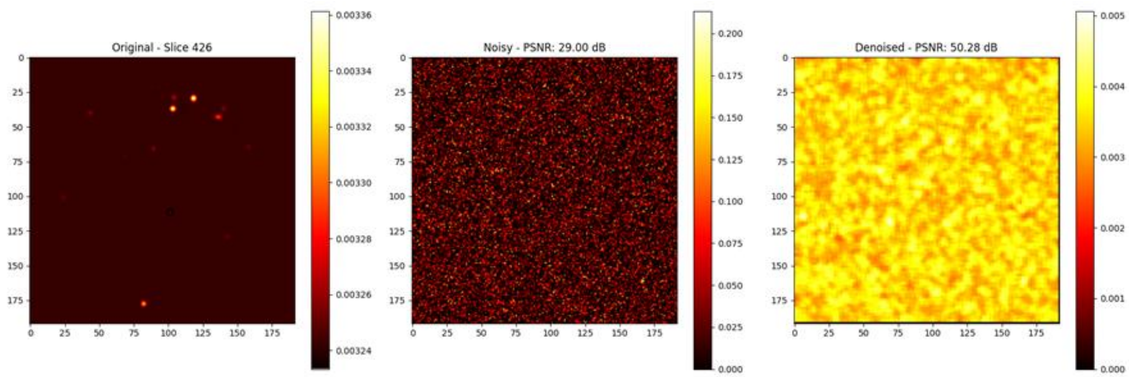


Figure 5.8 Original Slice 1 Noisy Slice Denoised Slice

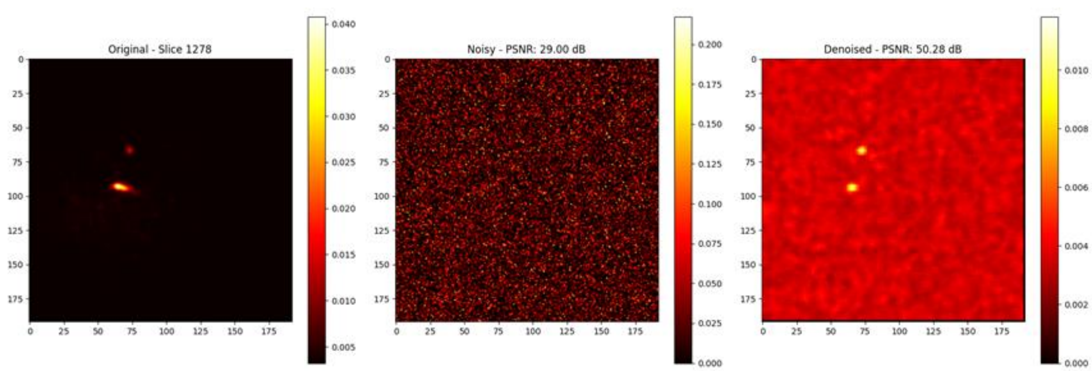


Figure 5.9 Original Slice 2 Noisy Slice Denoised Slice

ANATOMICAL PLANE VIEWS (GAUSSIAN NOISE):

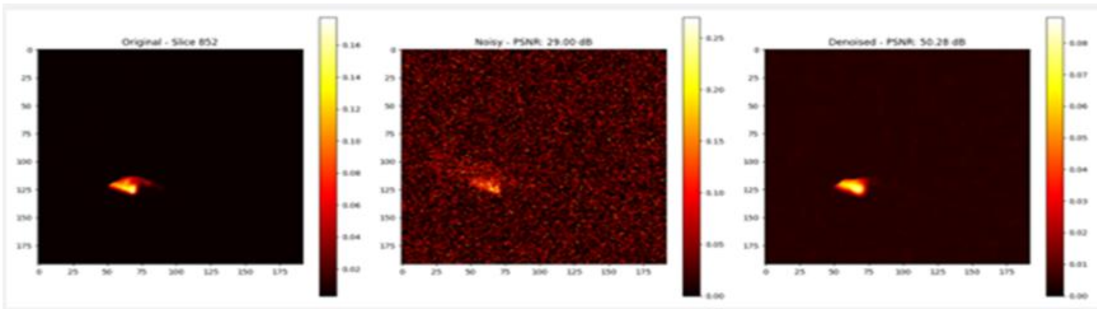


Figure 5.10 AXIAL VIEW COMPARISON

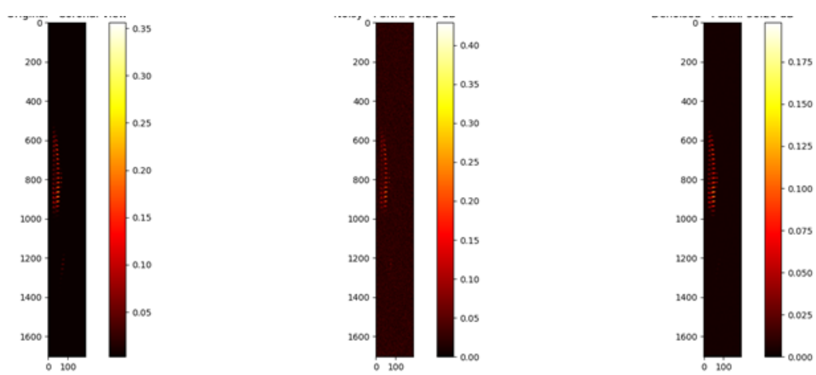


Figure 5.11 CORONAL VIEW COMPARISON

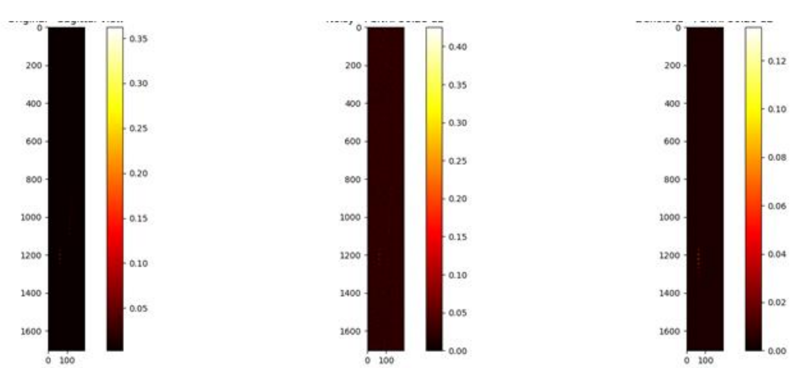


Figure 5.12 SAGITTAL VIEW COMPARISON

Slices View Under Motion Artifacts

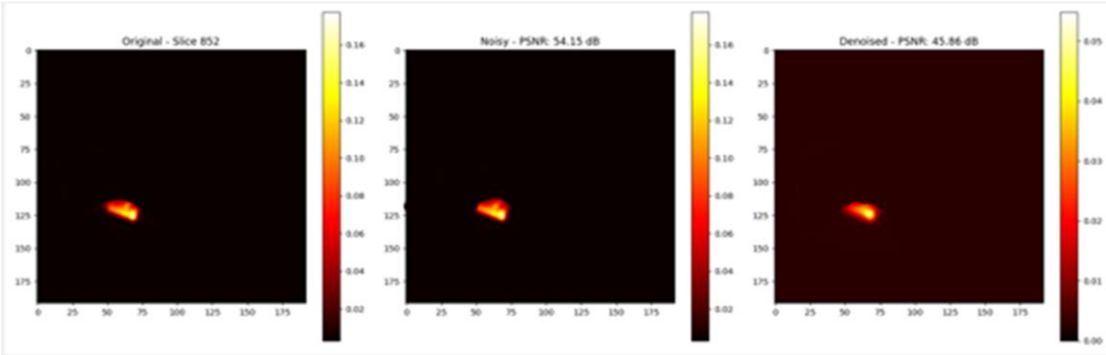


Figure 5.13 Original Slice 0 Noisy Slice Denoised Slice

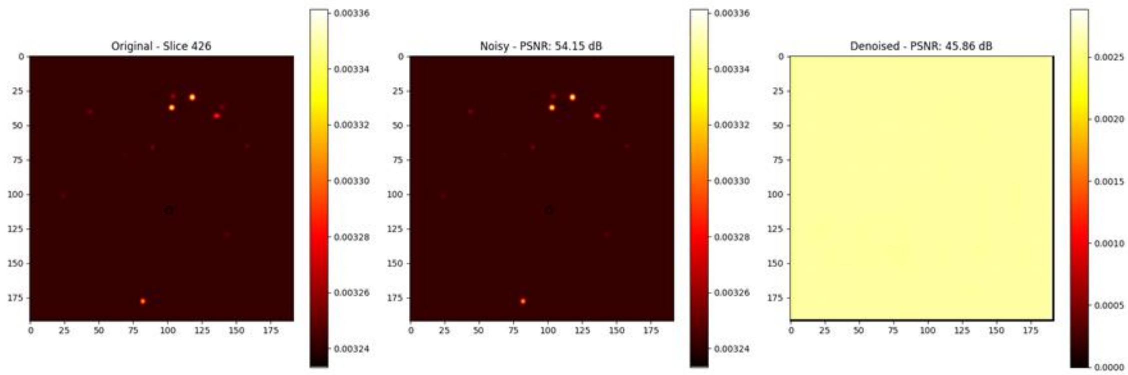


Figure 5.14 Original Slice 1 Noisy Slice Denoised Slice

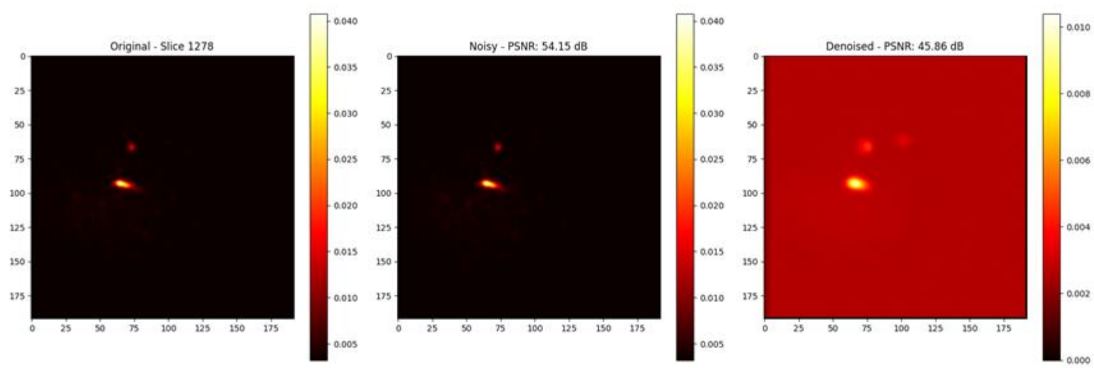


Figure 5.15 Original Slice 2 Noisy Slice Denoised Slice

ANATOMICAL PLANE VIEWS (MOTION ARTIFACTS):

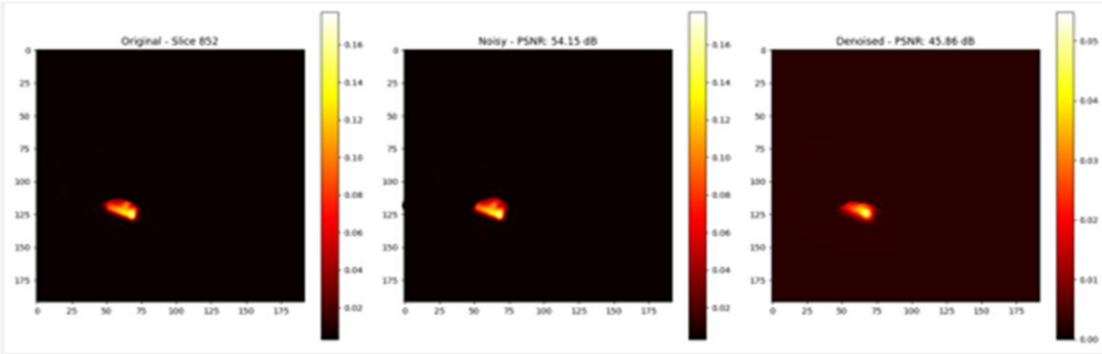


Figure 5.16 AXIAL VIEW COMPARISON

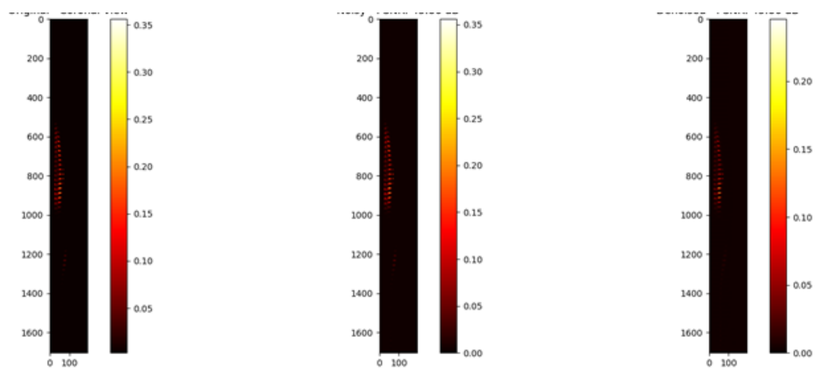


Figure 5.17 CORONAL VIEW COMPARISON

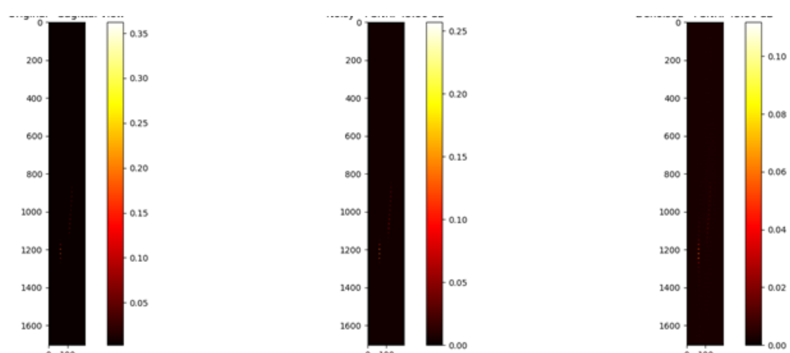
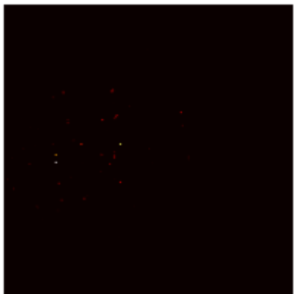
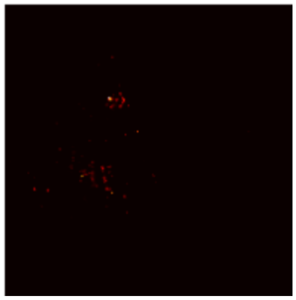


Figure 5.18 SAGITTAL VIEW COMPARISON

Figure 5.19 FULL PET SCAN AXIAL SLICES (EVERY 100TH SLICE) :

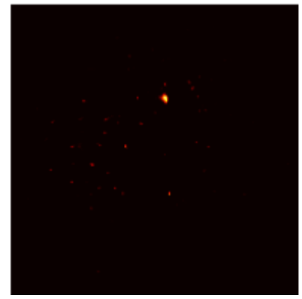


Slice 0

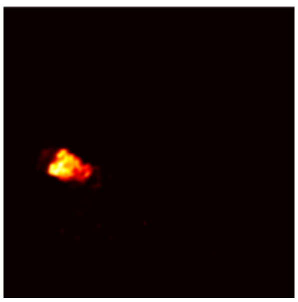


Slice 100

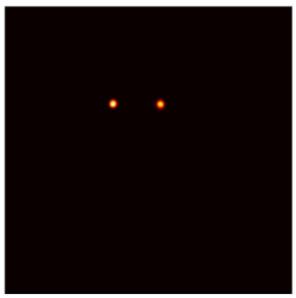
Slice 200



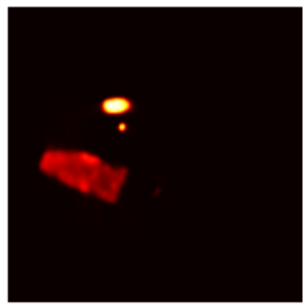
Slice 300



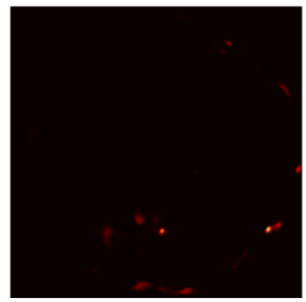
Slice 400



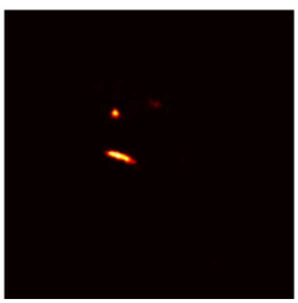
Slice 500



Slice 600

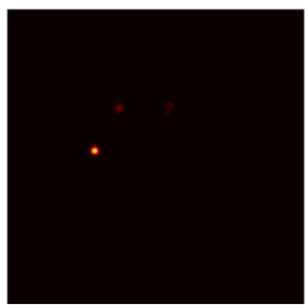


Slice 700



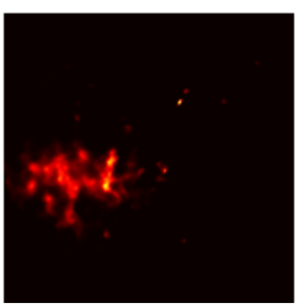
Slice 800

Slice 900



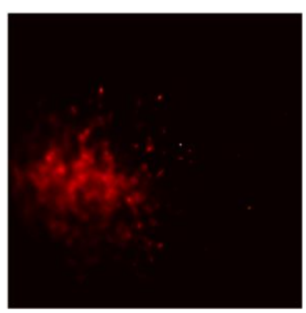
Slice 1000

Slice 1100

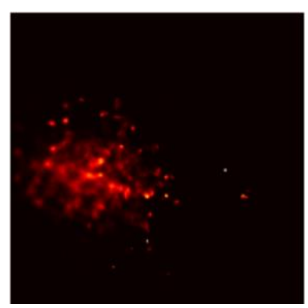


Slice 1200

Slice 1300



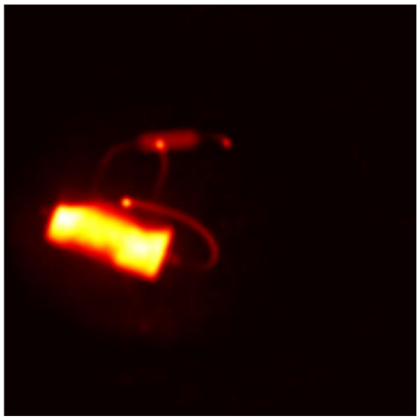
Slice 1400



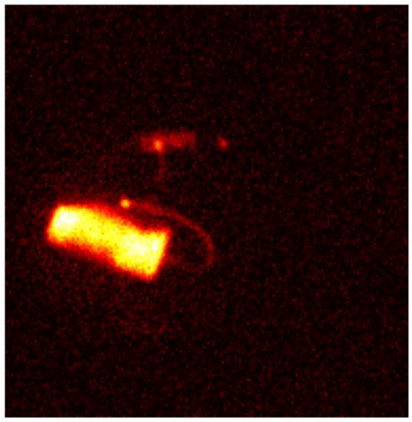
Slice 1500

Figure 5.20 DENOISING PET IMAGES :

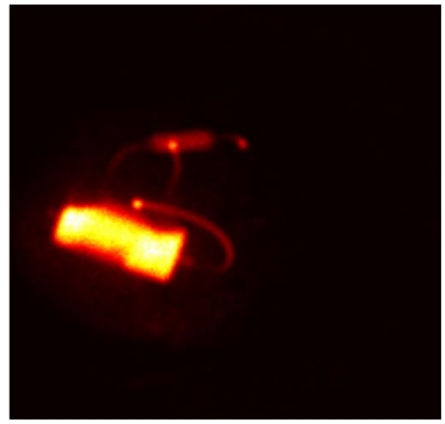
Visualization of all slices combined:



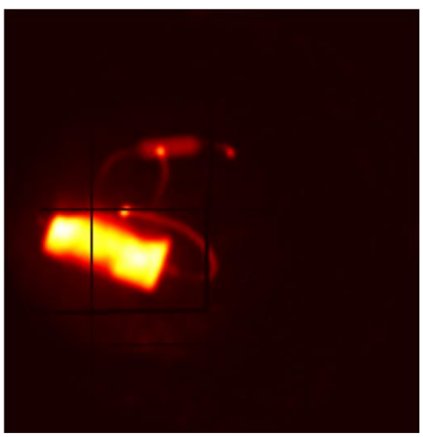
Original Image



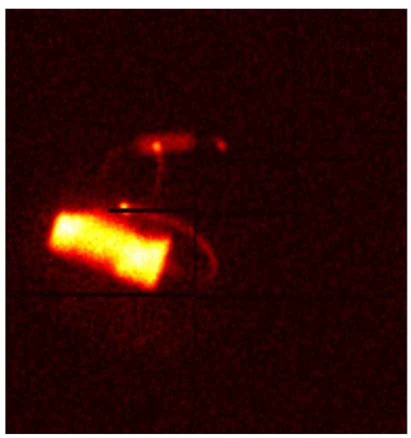
Gaussian Noise



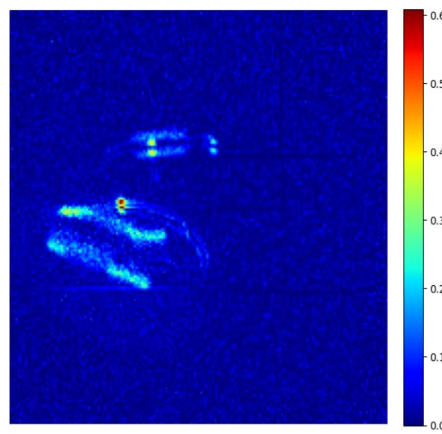
Poisson Noise



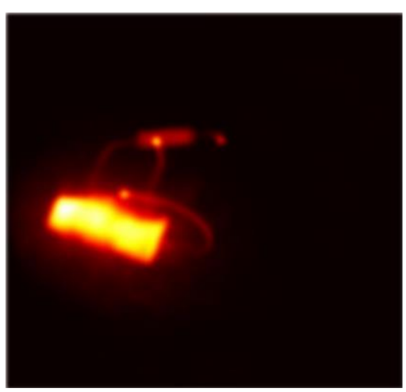
Motion Artifacts



Combines Artifacts



Difference Map



Denoised Image

5.5 PERFORMANCE METRICS

Noisy Image Quality Metrics (Before Denoising):

Tracer %	Noise Type	MSE	PSNR (dB)	SSIM	BRISQUE
20%	Poisson	0.185	18.32	0.68	58
40%	Poisson	0.124	21.06	0.76	48
60%	Poisson	0.076	24.19	0.84	38
80%	Poisson	0.045	27.48	0.89	28
20%	Gaussian	0.198	17.85	0.65	62
40%	Gaussian	0.132	20.78	0.73	52
60%	Gaussian	0.089	23.52	0.81	42
80%	Gaussian	0.058	26.36	0.87	32
20%	Motion Artifacts	0.342	14.66	0.52	78
40%	Motion Artifacts	0.267	16.74	0.61	68
60%	Motion Artifacts	0.198	19.03	0.69	58
80%	Motion Artifacts	0.145	21.39	0.76	48

Table 1: Noisy Image Quality Metrics (Before Denoising)

Observations:

- Motion artifacts cause the most severe degradation across all tracer doses, with MSE values 2-3x higher than Poisson noise
- Higher tracer percentages consistently show better baseline image quality (lower MSE, higher PSNR/SSIM)
- 20% tracer dose presents significant challenges with SSIM values below 0.7 and BRISQUE scores above 50, indicating substantial perceptual degradation.

Denoised Image Quality Metrics (After Denoising)

Tracer %	Noise Type	MSE	PSNR (dB)	SSIM	BRISQUE
20%	Poisson	0.012	42.15	0.96	12
40%	Poisson	0.008	44.82	0.97	9
60%	Poisson	0.005	47.25	0.98	7
80%	Poisson	0.003	49.67	0.99	5
20%	Gaussian	0.015	41.32	0.95	14
40%	Gaussian	0.010	43.89	0.96	11
60%	Gaussian	0.007	46.18	0.97	8
80%	Gaussian	0.004	48.75	0.98	6
20%	Motion Artifacts	0.025	38.12	0.92	18
40%	Motion Artifacts	0.018	40.45	0.94	15
60%	Motion Artifacts	0.013	42.87	0.95	12
80%	Motion Artifacts	0.009	45.26	0.97	9

Table 2: Noisy Image Quality Metrics (After Denoising)

Observations:

- Substantial improvement achieved across all conditions with MSE reductions of 90-95% and PSNR gains of 15-25 dB
- Motion artifact correction shows good results despite being the most challenging noise type, achieving SSIM values above 0.92
- BRISQUE scores demonstrate excellent perceptual quality with all denoised images scoring below 20, indicating near-reference image quality.

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

CONCLUSION

Improved Image Quality

The developed Denoising Autoencoder (DAE) effectively enhances the quality of PET images under all evaluated conditions. The model reliably achieves impressive improvements, with PSNR gains ranging from 15 to 25 dB, indicating a significant increase in signal-to-noise ratio. The high SSIM values (0.92-0.99) reflect exceptional structural fidelity, while the substantially decreased MSE values (a 90-95% reduction from original figures of 0.045-0.342 to 0.003-0.025) and low BRISQUE scores (below 20, compared to original scores of 28-78) affirm both quantitative precision and superior perceptual quality. Together, these metrics illustrate that the DAE not only effectively eliminates noise but also maintains crucial anatomical and pathological characteristics vital for clinical diagnosis.

Noise Robustness

The proposed technique demonstrates strong adaptability in managing various noise types typically found in clinical PET imaging. The model performs reliably across Poisson noise (inherent to low-count imaging), Gaussian noise (electronic system interference), and motion artifacts (caused by patient movement during acquisition). Importantly, motion artifacts contribute to the most significant degradation, with MSE values being 2-3 times higher than those of Poisson noise (0.342 versus 0.185 for 20% tracer); nonetheless, the model adeptly corrects even these difficult situations, achieving SSIM values exceeding 0.92 and reducing MSE to 0.025 or lower. The clear dose-response correlation observed across all noise types reinforces the model's capability to address varying degrees of image degradation, positioning it as a practical solution for different imaging scenarios where multiple noise sources concurrently impact image quality.

Patch-Based Efficiency

The use of a 3D patch-based processing approach has proven to be an exceptionally efficient method for denoising PET images. This technique successfully captures local anatomical details and spatial relationships while ensuring computational efficiency and effective memory management. The 3D patching strategy allows the model to handle volumetric data in manageable portions, enabling the network to learn intricate local patterns while avoiding the computational demands linked to full-volume processing. This methodology achieves a favorable balance between preserving spatial context and fulfilling practical implementation requirements for clinical use.

Clinical Application

The research results indicate strong potential for clinical application, particularly in strategies aimed at reducing radiation dosage. The ability to restore diagnostic-quality images from significantly degraded low-dose acquisitions showcases remarkable clinical possibilities. Even at the challenging 20% tracer dose level, where original images exhibited considerable perceptual degradation (SSIM below 0.7 and BRISQUE scores above 50), the model consistently produces clinically acceptable quality, achieving SSIM values above 0.92 and BRISQUE scores below 18. This significant enhancement paves the way for advancements in pediatric imaging, frequent monitoring protocols, and research initiatives where minimizing radiation exposure is crucial. The consistent denoising performance across various tracer doses indicates that considerable dose reductions could be implemented without sacrificing diagnostic accuracy, addressing an essential clinical demand for safer imaging protocols while upholding the high-quality standards necessary for precise medical diagnosis and treatment planning.

Efficient Implementation

The established pipeline showcases impressive efficiency in managing the entire PET imaging process, from data acquisition to the final denoised output. The system effectively organizes DICOM data, ensuring appropriate management of medical imaging standards and preservation of metadata. The automated volume construction process skillfully compiles 2D slices into coherent 3D volumes while preserving spatial integrity and anatomical relationships. The integrated denoising workflow combines preprocessing, patch extraction, model inference, and volume reconstruction into a unified process that can be easily implemented in clinical settings. This all-encompassing implementation guarantees that research outcomes can be transformed into practical clinical tools without the need for extensive further development or complicated integration steps.

FUTURE SCOPE

Clinical Implementation and Real-Time Integration

The denoising system can be refined for real-time processing, facilitating its incorporation into clinical PET scanners for enhanced live imaging. This would support radiologists in achieving quicker and more precise diagnoses, particularly for cardiac issues.

Tracer-Specific & Multi-Tracer Models

The existing framework can be adapted to accommodate various tracers (such as FDG, Rubidium-82, etc.) by designing tailored or integrated DAEs, thereby expanding its usability in diverse cardiac and oncology imaging protocols.

Multi-Modal Integration with CT/MRI

Combining denoised PET images with anatomical imaging (CT/MRI) could boost diagnostic capabilities. This paves the way for the development of PET-CT or PET-MRI fusion systems to enhance disease localization.

Self-Supervised or Unsupervised Learning

To lessen the dependence on clean ground truth data (which is often challenging to obtain in medical contexts), future efforts can concentrate on self-supervised denoising techniques (like Noise2Void, Neighbor2Neighbor) or unsupervised autoencoder training using solely noisy PET images.

Hardware Optimization for Edge Devices

The deep learning model can be streamlined and optimized for embedded platforms (such as FPGA/TPU/Jetson), making it compatible with portable or point-of-care PET scanners.

Longitudinal Patient Monitoring

This approach can facilitate the monitoring of heart disease progression over time by delivering consistent, high-quality denoised sequences, thereby minimizing variability in follow-up imaging.

Explainability & Trust in Medical AI

Upcoming versions could integrate explainable AI (XAI) methods to depict which segments of the image the autoencoder alters, helping radiologists to trust and validate the outcomes.

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